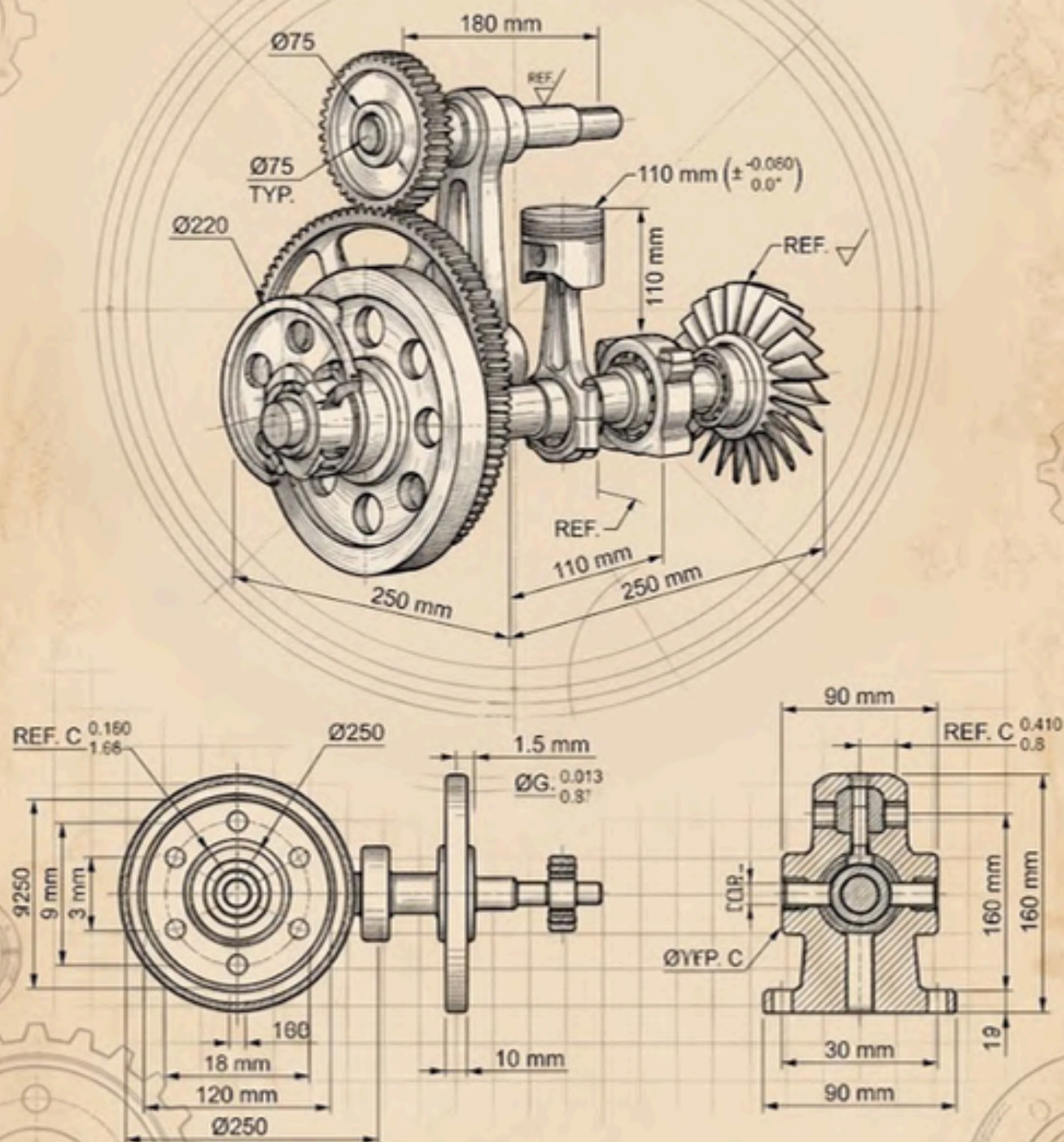


IMPULSE

Every great machine starts
with a single impulse



DEPARTMENT OF MECHANICAL ENGINEERING
KALYANI GOVERNMENT ENGINEERING COLLEGE

DEPARTMENT OF MECHANICAL ENGINEERING

mourns the demise of



Mr. Apurba Biswas

With profound grief and heavy hearts, the students and faculty mourn the sad demise of our beloved Apurba Biswas Sir. His unparalleled dedication, gentle kindness, and ever-helpful nature have shaped countless lives and will forever remain etched in our hearts.

We pray for his eternal peace and extend our deepest condolences to his family.

Rest in Peace

FROM THE DESK OF HOD

It gives me immense pride to unveil our brand new departmental magazine Impulse, a platform dedicated to showcasing the vibrant creativity, technical insight, and intellectual curiosity of our students, staffs and faculty members of the Mechanical Engineering Department. Education extends far beyond the confines of textbooks and classrooms, and this publication is a beautiful reflection of your diverse talents, perspectives, and voices. As the Head of Department, I have always believed in fostering the holistic development of our young minds. This magazine serves as a canvas for your innovative ideas, literary expressions, and artistic endeavors. It is a testament to the collaborative spirit and the relentless zeal of our student. I extend my heartfelt congratulations to the students, staff and faculty members, and everyone who contributed their time, energy and brilliant ideas to bring this vision to life. I hope this inaugural edition inspires you to think critically, communicate effectively, and continue pushing the boundaries of your imagination. I encourage all of you to actively participate, share your stories, and make this a regular and cherished tradition of our department. Wishing you all the very best in your academic and creative pursuits.

Warm regards,

Dr. Tridibesh Das
Associate Professor & Head,
Department of Mechanical Engineering
Kalyani Government Engineering College



FROM THE DESK OF REGISTRAR

It is indeed a good move by the Mechanical Engineering community of Kalyani Govt. Engineering College, Kalyani to start publishing a magazine by it. Being basically a Professor of Mechanical Engineering, I heartily welcome publication of this magazine first time in the history of KGEC, Kalyani from the Department of Mechanical Engineering. Department of Mechanical Engineering was started in the year of 1995 in Kalyani Engineering College, Kalyani (this was the name of college at that time). Then B.Tech (Mechanical Engineering) course was offered by the department with an intake of 30 students. Later in the year of 2003, the department started the 1st M.Tech course in this college in Production Engineering. In 2003–2004 session, this master degree course was conducted in collaboration with Central Tool Room and Training Centre (CTTC), Bon Hooghly, 24 Parganas giving thrust to Tool Engineering. However, this collaboration ended after one year, and with a revised syllabus, the department started conducting the course with its own faculty members and technical man power, and till today, myself is acting as this Coordinator of this M.Tech course. I am happy to state that on the whole, both the B.Tech (ME) and M.Tech (PE) courses are running in full swing under a good academic ambience. After the COVID-19 lock-down condition, there has been some challenges on the Teaching-Learning process in the department. This should be overcome at the earliest to raise the quality of regular Teaching-Learning process in the department. Along with running the PG course and having most of the faculty members obtained Ph.D degree, the department has also conducted several Ph.D programmes supervised by faculty members of the department. In some cases, good collaboration also exists with different departments of the college and other institute/organization of repute w.r.t. the works related to Ph.D programme, M.Tech dissertation and also B.Tech projects. These works lead to good quality research publication also. I personally wish the department of Mechanical Engineering should take lead role to get more and more involved in the Teaching-Learning process and research activities in upgrading the department by overcoming different imminent challenges. Mechanical Engineering Department also takes a lead role in Students' Automobile Club of Kalyani Govt. Engineering College (SAC-KGEC) related activities involving students and interested faculty members of all the departments of college. Through designing and making new GoKarts, participants acquire good amount of practical in-hand experience to shape them as a good engineer. Two SAC-KGEC teams also participated in all India competitions in the year of 2024 and 2025 held at Coimbatore, Tamil Nadu exhibiting their tremendous will power and zeal. Hope with the active participation of all persons involved, publishing Mechanical Engineering Department magazine will be successful, and it will continue in each year to come with the same zeal and vigour.



Prof. Dr. Santanu Das
Professor & Former Head, Mech. Engg. Dept. and Registrar (Offg)



COMMITTEE MEMBERS

4th Years

AFZAL HOSSAIN MALLICK

3rd Years

ARITRA MUKHOPADHYAY

ARPAN CHATTOPADHYAY

ANKUR BHATTACHARYA

SAHIL SHEIKH

2nd Years

MD IMRUL HOSSAIN

DEBRAJ CHAKRABORTY

NISARGA PAL

SUBHARTHI SARKAR



অবুঝ

এই ঝড়ে

থাকবো না আর বন্ধ ঘরে
ভিজব জবর পড়ব জ্বরে
ভাঙব কোমর পিছলে পড়ে-
নেই পরোয়া কুছ!
গাছপালারা মাতলো নেশায়
প্রাণের জোয়ার মাঝদরিয়ায়-
মনমাঝি অবুঝ!

- ড. দেবজ্যোতি মিত্র

অধ্যাপক, মেকানিকাল ইঞ্জিনিয়ারিং বিভাগ



Under the crimson state
Subham Sau
ME, 2025

অপেক্ষা

সদ্য বসন্ত।

বিকেলবেলা।

গাছের চূড়াগুলো

সোনালী ফাগ মেখেছে

এই স'বে।

একটা মৃদু হাওয়া -

না গরম না ঠান্ডা।

মাঠের মাঝের

সরু রাস্তাটায়

বড্ড স্পষ্ট -

পায়ের চিহ্ন।

একটা কোকিল -

গলা ভেঙে গেছে,

তা'ও ডাকছে।

কয়টা শালিখ

ধুলো মাখতে মাখতে

গল্পে মশগুল।

তাই না দেখে

দুটো বাচ্ছা

সোনা গায়ে মেখে

খেলতে নামল মাঠে।

একটা পুরনো গন্ধ -

অনেকটা নতুনের মত,

নতুন করে ভাবতে শেখালো

তোমার কথা।

ঠিক এমনই এক দিন -

তুমি আসবে বলেছিলে।

- ড. দেবজ্যোতি মিত্র

অধ্যাপক, মেকানিকাল ইঞ্জিনিয়ারিং বিভাগ

এইতো সেই কিছুদিন আগের কথা।

কিছু মাস বছর, ঠিক মনে পড়ছে না,
কেবল মনের ভিতর দোল খাচ্ছে সেই কচি পাতার নাচ।
গাছটাতে পাতা খুব একটা ছিল না
কিন্তু যেগুলো ছিল সবুজ ছিল, সতেজ ছিল।
প্রকৃতির নিয়মে আপন খেলায় থাকত মেতে।
দিনের আলোয় ছুটে বেড়াত, অন্ধকারে পড়ত ঘুমিয়ে।
আবার ভোরের ওই পাখির ডাকে উঠত জেগে।
প্রভাতের লাল টকটকে রবির দিকে হাঁ করে থাকত।
কিন্তু, হঠাৎ কে যেন তার কানে ফিসফিস করে বলল -
'আরে এমন আছিস কেন? বড় হয়ে যা।
এসব আলোর সাথে, বাতাসের সাথে খেলা করলে চলবে?

ধীরে ধীরে যদিও পাতারা বাড়তে লাগল।
কচি সবুজে বয়সের আভা এলো।
বাতাস এখন ডাকতে আসলেও সে চুপ থাকে।
সূর্যের সাথেও হয়েছে আড়ি।
সেই কচি ছোট ডালটা হয়েছে রুম্ব, মোটা।
কতকগুলো উই ছাড়া কেউ থাকে না সেখানে।
যদিও পাখিরা অস্তে চায়,
কিন্তু পাতাগুলো চঁচিয়ে বলে, "আমায় বড় হতে দে।
চলে যা তোরা।"

জানি না এই বড় হবার পরিণতি কী?
এমন রাগী, খিটখিটে করে বসে থাকা কেন?
কার জন্যে অপেক্ষা করে আছে?
যার জন্যে আসে সে কি আসবে?
এলেও তাতে এতটা ওজন আছে কি
যে দীর্ঘ এতো অপেক্ষার বাটখারা চাপানো যায়?
মাছ জলের বাইরে অপেক্ষা করতে এলে যেমন করে,
ঠিক তেমনি চটপট করছে সে।
তবু ঝাঁপ দিচ্ছে না।
কার অপেক্ষা করছে সে?
পাতাগুলো বড় হতে গিয়ে বুড়ো হয়ে যায়নি তো?

- Arpan Chattopadhyay
ME, 3rd Year



- Nisarga Pal
ME, 2nd Year

Biology ও ইঞ্জিনিয়ারিং

জীবনের গোপন সুরে Biology কথা কয়,
প্রকৃতির নীরব বাণী জ্ঞানের দ্বারে রয়।

পাতার শিরায়, ফুলের হাসি, জীবনেই গান,
সেই সুর নিয়ে ইঞ্জিনিয়ার গড়ে নতুন প্রাণ।

যন্ত্রের মাঝে খুঁজে ফেরে প্রকৃতির পরিচয়,
জীবনের এই রহস্যতে ভবিষ্যতের জয়।

মানবসেবার পথে পথে জ্ঞানের দীপ জ্বলে,
Biology আর প্রযুক্তি মেশে কর্মফলে।

প্রকৃতি আর বিজ্ঞানের মিলন-সুরের তান,
তাহার বুকে গড়ে ওঠে আগামী অন্বেষণ।

- Akash Pramanik
ME, 2nd Year

গহীনের লেখা

এঁরে তখন পড়ন্ত বেলার রোদ; বহু বছর পূর্বে শহরের বুক চিরে এগিয়ে চলা ট্রামের নিষ্প্রভ আলোয় কেউ কোন একদিন শক্তি চট্টোপাধ্যায় পড়েছিলেন, মুগ্ধ হয়েছিলেন। সেই মুগ্ধতার রেশ ধরে আমরা অনেকটা পথ হেঁটেছি, ফুটপাথ জুড়ে। আমরা তখন শক্তি পড়ে "পা থেকে মাথা পর্যন্ত টলমল করে"; "ফুটপাথ বদল হয় মধ্যরাতে", অচিরেই। সে এক দিন ছিল বটে!!

দীর্ঘদিনের অনভ্যাস কাটিয়ে আজও আবার ফুটপাথে আছি, সাথে কেউ নেই, আছি আমি একলা। এই ফুটপাথ ধরে হাঁটতে হাঁটতেই আমরা প্রুফকের 'লাভ সং' শুনি; সমস্ত ইনহিবিশন কাটিয়ে ওয়েস্ট ল্যান্ড থেকে জেগে ওঠা স্বপ্নেরা আমার খসরু থেকে রবীন্দ্রনাথ ছুঁয়ে ওমর খৈয়ামে বুদ্ধ হয়ে থাকে, সন্ত কবীরের হাত ধরে "জীবন দেবতা"র খোঁজ করে, প্রিয়পরমের প্রীতির কারণ হয়ে ওঠার চর্যায় নিয়োজিত থাকে, পথ হাঁটে, হাঁটতে থাকে অনন্তের পথে। হাঁটতে হাঁটতে আরশিনগরের দিকে যায়। পড়শীর খোঁজ করে।

আর, তখন শহরের নিয়ন আলোয় ভিজে থাকা ব্যস্ত ফুটপাথেই জুড়ে যাচ্ছে মানুষের সাথে মানুষের যন্ত্রণা, মিশে যাচ্ছে মানুষের হাসিমুখ, তাদের ব্যথার সাথে সখ্য জমে উঠছে। এভাবেই, নিঃশব্দে কখন যেন এই গহীন মধ্যরাত একরাশ স্বপ্ন হয়ে উঠছে!

- Subharthi Sarkar
ME, 2nd Year

আমাদের কলেজ জীবন

ভোরের আলো মেখে শুরু নতুন দিন,
হাসির শব্দ ভরে ওঠে কলেজের ক্যান্টিন।

ক্লাসরুমে স্বপ্নগুলো ডানা মেলে যায়,
বন্ধুত্বের গল্পগুলো হৃদয় ছুঁতে চায়।

ক্যান্টিনের কোণে বসে আড্ডার আসর,
চায়ের কাপে ভরে ওঠে কত সুখের খবর।

শেষ বেঞ্চ জানে কত অজানানো কথা,
হাসতে হাসতেই কেটে যায় দিনের ব্যস্ততা।

প্র্যাকটিক্যালের ভয়, পরীক্ষার চাপ,
তবু বন্ধু পাশে থাকলে মুছে যায় সব অভাব।

ফেস্টের রঙিন আলো, গিটারের সুর,
এই ছোট মুহূর্তগুলোই করে জীবন ভরপুর।

বৃষ্টিভেজা পথে পাশাপাশি হাঁটা,
কারও চোখে হয়তো নিজের গল্পটা দেখা।

হয়তো বলা হয় না মনের সব কথা,
তবু কিছু অনুভব রয়ে যায় দিয়ে নীরব ব্যথা।

কেউ এখানে খুঁজে পায় নিজের পরিচয়,
কেউ স্বপ্ন বুনে রেখে বিদায়।

আবার কেউ অজান্তেই খুঁজে পায় এমন একজন,
যার হাসিতে ভালো লাগে ব্যস্ত এই জীবন।

KGEC শুধু নাম নয়,
এ যেন এক টুকরো অনুভব, এক মায়াময় সময়।

এখানে রাগ আছে, অভিমান আছে,
আছে একসাথে পথ চলার অদৃশ্য বাঁধনা।

একদিন হয়তো সবাই চলে যাব দূরে,
সময়ের স্রোতে, ব্যস্ত শহরের ভিড়ে।

তবুও মনে পড়বে এই সবুজ ক্যাম্পাসখানি,
যেখানে হাসির শব্দে মুখরিত ছিল জীবনের বাণী।

বহু বছর পর ক্লান্ত কোনো বিকেলে,
হয়তো হঠাৎ স্মৃতি ফিরে আসবে অবহেলে।

মনে হবে,
জীবনের সবচেয়ে সুন্দর অধ্যায় ছিল সেই সময়,
যখন “আমরা” শব্দটা
এতটা আপন ছিল সবার হৃদয়।

– Avrajit Dey
ME, 1st Year



– Racktim Das
ME, 3rd Year

বর্ষা শেষে

বর্ষা শেষে সূর্য হাসে, আকাশ ধরে রঙ।

রামধনু ঐগেছে পড়ে,

দিনের আলোয় মুক্তো ঝরে—

কারলাগি যে বসে থাকিস্ সেজে তুই মা সঙ।।

নৃত্য করে ভেকের দল, জল ছুটছে কলকল।

নদীর বুকে বান এসেছে, পুকুর নালা ভরে গেছে—ডাঙা দিয়ে ছুটছে কৈ জিওল মাছের দল।

কিসের জন্য সঙ সেজেছিস—কারলাগি তুই বল।।

কেন এমন রূপ ধরেছিস, তোরে কে সাজালো বল!

বলদ জোড়া সামনে করি—যুবক লাঙ্গল কাঁধে ধরি,

মাঠের কাজে সারি ধরে যাচ্ছে কৃষকদল।

কার লাগি তুই সাজলি মাগো—কিসে মজলি বল।।

গাছের ডালে ফিঁঙে ডাকে মাঝে মাঝে সূর্যি টাকে —জলে ছুটে মরালী সব, নদীর চরে বক—

কে তোরে মা করবে বরন?

মনে কি তোর ছক!

বৃষ্টি জলে সিক্ত মাটি, চাষের জন্য হয়গো খাঁটি —

ধানের বীজ বুনবে কৃষক —সাজছে চাষীরদল।

কার তরে মা মন মজালি, কেন সাজলি বল!

রাজহংস জলে খেলে, মাছরাঙাটি ঝাঁপে জলে—

কচি ঘাসে হাওয়া খেলে পাতায় ধরে রঙ।

যাবি নাকি মা শ্বশুর বাড়ী?

তাই কি সাজলি সঙ?

আম-কাঁঠালের অঘোর বনে,

দোয়েল বসে আপন মনে—

মাছরাঙা যে ছুটে বেড়ায় মাছটি ধরার ছল।

সবুজ রঙের পরলি শাড়ী, কেন রে মা তুই বল।

সিন্দুর পরলি দিনের শেষে,

ঘোমটা নেইকো সূর্য্যদেশে —

অমন উদার মুক্তবেনী কোথায় পেলি বল।

ঘোমটা ছেড়ে পরলি সিন্দুর কেমনি এ তোর ছল।

কিসের তরে সাজলি মা তুই কেনই এমন রঙ—

কে তোরে যে করবে বরণ, দেখে এমন সঙ।

- সুশান্ত বারিক

অধ্যাপক, মেকানিকাল ইঞ্জিনিয়ারিং বিভাগ

অন্ন

আজ অনেক কাল হইলো,
আমার শরীর ফাঁকা,
ক্ষুধায় আমার শুকনো নাড়ি,
মৃত্যু ছবি আঁকা।
হাত পেতে তাই চাইছি
আমায়, অন্ন দাও দুটি;
তাহার স্বাদ লইয়া আমি,
আবার জেগে উঠি।
দেয়নি দিতে অন্ন তাহারা,
বাঁধলো তাহার হাত,
যেই হাতেতে অন্ন ছিল,
দিল সেই হাতের আঘাত।
সাহস কত ভিক্ষুক তুই!
অন্ন খেতে চাস,
হাতের দিকে আর যদি চাস,
তোর, দেবো গলায় ফাঁস।
চাইনি আমি অন্ন খেতে,
বরেছে তরল দেহের;
ক্ষুধা আমার অন্নের নয়,
ক্ষুধা তাহার প্রেমের।
অন্ন আমার কাছেই আছে,
খাওয়ার লবণ টুকু নাই;
তাই চোখের জলের সাথে আমি,
অন্ন মেখে খাই।।

- Souvick Ray
ME, 1st Year

বাড়ি ফেরা

আজ বাড়ি ফেরার দিন।
অনেকদিন পাইনি দেখতে চেনা প্রকৃতির মুখ,
পাইনি খেতে মায়ের হাতের গরম ভাতের সুখ।
ঘুমটা আজ ভাঙলো ভোরে, ডাকতে হয়নি কারও ;
বাড়ি যাব বলে কথা, ঘুমের গুলি মারো।
সকাল সকাল না খেয়ে তাই ছুটলাম স্টেশনে,
বাবার ফোন আসতে বললাম "উঠে গেছি ট্রেনে"
আহা কত আনন্দ আজ আমি বাড়ি যাব!
গরম ভাত, মন্ডা - মিঠাই গিয়েপ্লরে খাবো।
সকলকে দেখব একসাথে, মনে মাতবে সুখ মেলা
করব ভাই - বোনের সাথে সারাটা দিন খেলা।
দেখতে দেখতে চলে এলাম চেনা পথ ধরে,
ঘুমটা আজও ভেঙে গেল হঠাৎ করে ভোরে।
বিছানাতে শুয়েই গলা হয়ে গেল ভারী,
দুদিন কেটে গেল এবার আবার দেবো পাড়ি।
আচ্ছা তবে চলি,সময় হয়ে এল ক্ষীণ,
হেসে বলি,আসবে আবার 'বাড়ি ফেরার দিন'।

- Souvick Ray
ME, 1st Year



- Sudipto Sardar
ME, 1st Year

বিকৃত

আমি এই বিশ্বচরাচরে কতবার মরে
গেছি খোঁজ রাখনি কেউ।
আমার অন্তরাত্মার থেকে এখন
ধুলোবালির ক্যাথেরসিস।
আমি মুষ্টিবদ্ধ হাতে আমার বাহ্যিক
চেহারার ডিসক্রেপন্সি কে সরাতে
বদ্ধপরিকর।
শুধু নেতিবাচক সংসর্গ আজ আমায়
ইতিবাচক এর অর্থানুভূতি জাগিয়েছে।
হাহাহাহা
হাহাহাহা হাহাহাহাহা !!!!!
বড়োই অদ্ভুত আমার চরাচর, যেখানে
আপন হয় পর,
তাই যারপরনাই করে নিজের
উদয়মানতাকে জারি রাখতে পর এর
দ্বারে বারেবারে হাত পাততে হয়।
হাহাহাহাহা হাহাহা হাহাহাহা!!!!!!
আজ আমার হৃদয়পুর এ হৃদয় এর
খেয়াল কেউ রাখনি তাই অনন্ত বিষাদ
এর ছাপ করছে আমাকে দেখে নিজেই
বিলাপ।
আবার প্রবুদ্ধতার হাসি।
হাহাহাহা হাহাহাহাহাহা !
বিকৃতময়

- Subharthi Sarkar
ME, 2nd Year

মেঘ বালিকা

নীল আকাশে ভাসে মেঘ বালিকা ধীর,
রূপালী আঁচল দোলে, সে যেন সুধীর।
বৃষ্টি হয়ে ঝরে, ভিজে যায় ধরণী সব,
আমার মনে তার ছবি, সে এক উৎসব।
বজ্র যখন কভু গর্জে ওঠে ঘোর,
মনটা আমার তখন হয় না শান্ত, নড়।
বিদ্যুতের চমকে দেখি তার রূপ,
আমার ভালোবাসা থাকে যেন চুপ।
দিনের শেষে যখন সে যায় মিলিয়ে,
তারার আলোয় তার স্মৃতি জাগে
বিলিয়ে।
তবুও জানি সে আছে আমারই পাশে,
প্রেমের পরশ নিয়ে বাতাসে ভাসে।

- Aahir Maji
ME, 4th Year



- Racktim Das
ME, 3rd Year

সময়ের মাপকাঠি

অপ্রাপ্তির নৌকা

ছি ল এক ঘড়ি, টিক-টিক তার প্রাণ,
সময় মেপে চলতো সে, গাইতো আলোর গান।
কিন্তু তার মালিক যে, বড়ই বে-হিসেবি,
সময় নিয়ে ভাবে না সে, যেন এক আজগুবি।

কেউ তাকে নাহি ডাকে, কারো কাছেই নেই মান,
সকাল-সন্ধ্যা সব তার কাছে সমান।
তাই তো ঘড়ি মুচকি হেসে কয়,
"আমার আবার ঘড়ির কিবা দরকার হয়?"

টিক-টিক করে দিন কেটে যায়, রাত আসে,
ব্যস্ত লোকের ভিড়ে আমি রহি শুধু পাশে।
আমার হাতে কাজ নেই, নেই কোন তাড়াহুড়ো,
সময় আমার সস্তা যে, হয় না কোনো খুঁড়ো।

- Aahir Maji
ME, 4th Year



- Nayan Biswas
ME, 2nd Year

নদীর তীরে দাঁড়িয়ে খুব মায়াবী একটি নৌকা...
নদীর ঠান্ডা মিষ্টি বাতাসে নৌকাটি, কী অপূর্ব
সেই দৃশ্য।
মাঝে মাঝে ভাবি নৌকায় চড়ে এক বন্ধুর পাড়ি
দেব। বন্ধুরবর.....

তা সে নদীতে যত বাধাবিপত্তিই আসুক না কেন।।
এই ভেবে মন যেন কেমন আকুল হয়ে ওঠে।।
কিন্তু নৌকায় সর্বদা বহু যাত্রীর ভিড়।
কিছু থাকে নিত্যযাত্রী আর আবার কখনো
কখনো কিছু নতুন যাত্রীর আগমন ঘটে।।
তারা প্রতিদিন নদীর এপার থেকে ওপার পাড়ি
দেয়। প্রতিদিন.....

পাড়ি দেওয়া হয়ে ওঠে না কেবল আমার।
অবশ্য ভুল টা আমারই।। আমাকে তো কেউ বাধা
দেয়নি নৌকায় চড়ে যেতে।
কেবল আমার একটি অন্যায় আবদার যে আমি
একা চড়ে পাড়ি দেব সেটা কি আর মানা
সম্ভব,
সম্ভব না, সত্যিই সম্ভব না।।

অথচ এর আগে অনেক সময় নৌকাটিকে একা
নদীর ওপর ভেসে থাকতে দেখেছি, পাশে গিয়ে
বসেওছি।
তখন হয়তো একা যাওয়া সম্ভব হতো।।
ভাবতাম পরে নাইয় কোনোদিন চড়ে নেব।
কিন্তু যত দিন যায় নৌকায় যাত্রীর ভিড় বাড়তেই
থাকে।
তাই ঐ নৌকায় একা চড়ে সাত সমুদ্র তেরো নদী
পারা পারের স্বপ্ন টা স্বপ্নই রয়ে গেল।।

-Debraj Chakraborty
ME, 2nd Year

ঐশ্বরিক

কাফনের আড়ালে

ধরেছিলো গাছে আমার তখন অন্য এক
ফুল-
সেই ফুলের নেই কোনো খুঁত,
বিশ্ব চরাচরে হন্যে হয়ে শুধু খুজছিলাম একুল
হয়ে ও - কুল।
জানি না পাবো নাকি সেই ঐশীক ফুল!
যা শুধু আমার হওয়ার কথা, হয়েছে অন্য
কারো,
প্রণয়ের দ্বারে লুটিয়েছি তার, গুনলে প্রায়
হাজার বারো।
প্রণয় থেকে পরিণয় হবে কি না জানা নেই তা!
সুধিয়েছি নিজেকে শুধু আশা করা এই
জীবনে শুধু খামোখা।
বনলতা সেন তুমি, বিশ্বাস করলাম তোমায়
আশ্রয় দিও তোমার এনাম্লির মণিকোঠায়।
আশ্রিত আমি, লুটিয়েছি তোমার পদতলে
করবে তুমি গণ্য?
আসেনা কি তোমার মনে প্রেম একরাশ
আমায় নিজের করে নেওয়ার জন্য?
যদি তুমি বলো "চাইনা তোমায় চাই শুধু
আমার অগ্নিহোত্রী,!"
তবে সরে যাবো আমি তোমার থেকে, হবো
চির বিনিদ্রযাত্রী।
নিদ্রার ঘোরে হয়তো আসবে তুমি, তবে
থাকবে তোমার মনে জিঘাঙ্কশা,
আমি কিন্তু করবোনা তবুও তোমার
অগ্নিহোত্রীর প্রতি হিংসা।
হয়তো আমি হতে পারিনি তোমার নিয়ত
যজ্ঞকারী,
তবুও কি একটি বার তোমার হাতটি ধরতে
পারি??

- Subharthi Sarkar
ME, 2nd Year

একদিন কাফনের আড়ালের মাটি হয়ে
হাজার বছর এই ধরায় থেকে যাবো
রয়ে।
বৃক্ষ জন্মাবে আমার ওপর, পাতা হবে
তাতে,
শিকড় গুলো আমার বুকের ভিতর
স্পর্শ করতে করতে
আঁকড়ে নেবে আমাকে।
ফোটা ফোটা বারি হবে সঞ্চয়,
আমার বৃক্ষ তলে বহু জীব নেবে
আশ্রয়।
উৎকৃষ্ট বিশালাকার বৃক্ষ হবে আমার,
ঝরে যাবে পাতা, গজাবে আবার
হাজার হাজার।।
আবারও শিকড় আমার বুকের ভিতর
স্পর্শ করতে করতে আঁকড়ে নেবে,
তোমাদের আশ্রয় দেবে,
দেবে ছায়া,
শুধু উপভোগ করতে পারবেনা আমার
অশরীরী এই কায়া।

- Subharthi Sarkar
ME, 2nd Year



- Sudipto Sardar
ME, 1st Year

সন্ধের আকাশে মেঘেদের মেলা

আকাশে আজ মেঘেদের মেলা,
উড়ে চলে যেন কোনো ভেলা।
রঙিন স্বপ্নে আঁকা ছবি,
ভেসে ওঠে ওই দূর রবি।

সেথা থেকে আসে এক মৃদু সুর,
মন ভোলায় সকল বিভোর।
কার হাতে যেন সেজে ওঠে নীড়,
সে তো আমার প্রিয় মেঘবালিকা স্থির।

আলোর রেখায় তার সাজানো পথ,
রূপকথার দেশে চলে তার রথ।
মুগ্ধ করে তার হাসির ধারায়,
আনন্দ যে মন ছুঁয়ে যায়।

আঁধারে জ্বলে তার আলোর শিখা,
প্রাণে আনে এক নতুন লিখা।
স্বপ্ন হয়ে সে আসে বারে বার,
আমার মন শুধু তারই দিবার।

মেঘবালিকা তুমি যে এক আশা,
ছড়িয়ে দাও শুধু ভালোবাসা।
তোমার ছোঁয়ায় জাগে প্রাণ,
তুমিই আমার সকল গান।

- Aahir Maji
ME, 4th Year

স্মৃতির ক্যাম্পাস

পুরোনো করিডোরগুলো আজও যেন আমাদের
গল্প,
কত হাসি, কত অভিমান, কত অজানা কথা।
চার বছরের পথচলা আজ স্মৃতির পাতায় বন্দী,
তবু প্রতিটি মুহূর্ত যেন এখনও জীবন্ত প্রতিধ্বনি।
ভুলিনি স্যার, SAC ওয়ার্কশপের সেই দিন,
শেষ সেমিনারে সবার মুখে উচ্ছ্বাসের রঙিন
স্বপ্নখানি।

ম্যাথেমেটিক্স, Solid Mechanics আর মেশিন
ডিজাইনের চাপ,
তবু বন্ধুদের আড্ডায় মিলতো ক্লাস্তির প্রতিষেধক।
হোস্টেলের রাত, এক প্লেট খাবার, হাজারো স্মৃতি,
রক্তের সম্পর্ক না হলেও বন্ধুত্ব ছিলো জীবনটা।
কঠিন সময়ে ছায়াসঙ্গী ছিল যে বন্ধুরা আর শক্তি,
তাদের সাথেই আঁকড়ে থাকবে জীবনের শ্রেষ্ঠ
স্মৃতি।

আজ আমরা পাশাপাশি, শুরু নতুন পথের যাত্রা,
স্বপ্নগুলো ভাসে চোখে, তবু অতীতের মায়া।
তবু KGEC-র সেইদিনের কথাগুলো হারবে না
শেষ,
বন্ধুত্ব আর স্মৃতিগুলো রয়ে যাবে সবার অন্তর
অবশেষ।

কলেজ শেষ হয়েছে, কিন্তু গল্পটা নয়,
বন্ধুত্ব এই পথের সূচনা কেবলই হারায় না।
সময়ের স্রোতে দূরে হারালেও একই রবে চির—
KGEC শুধু একটি কলেজ নয়, আমাদের
জীবনের শ্রেষ্ঠ স্মৃতি।

-Alok Majee
ME, 2025

মার্ক শিট

ছেলের মাধ্যমিকের মার্কশিটটা হাতে নিয়ে অর্ধেন্দুবাবুর চোখদুটো ছলছল করে উঠল। অঙ্ক, ভৌতবিজ্ঞান, জীবনবিজ্ঞানে ফুল মার্কস। শুধু স্কুলে নয়, ও তল্লাটে এরকম রেজাল্ট আগে কেউ করতে বলে কারো মনে পড়ে না। তবুও মাধ্যমিকের আগে ভাইরাল জ্বরে আক্রান্ত না হলে রনি হয়তো রাজ্যে একটা পজিশন অবশ্যই থাকত। মা মিনতিদেবীর চোখেমুখে বাঁধনহারা আনন্দ। ফোনে একের পর এক শুভেচ্ছা বার্তা আসছে। এইমাত্র রনির পাড়ার এক বন্ধু খবর দিয়ে গেল, রনি নাকি ডিস্ট্রিক্ট-এ ফার্স্ট হয়েছে। এত খুশি এত আনন্দের মাঝে রনি কিন্তু চুপচাপ। আগে এরকম ছিল না ও। বছর চারেক হল, রনির আবেগে আনন্দ যেন কোথাও গুম হয়ে গেছে। এমনই এক রেজাল্টের দিন ছিল সেবার। উচ্চমাধ্যমিকের রেজাল্ট। সেবার রনির উৎসাহ ছিল চোখে পড়ার মতো। ক্লাস সিক্সের রনি ওর দিদির রেজাল্টের জন্য সকাল থেকে ভীষণ এক্সট্রাইটেড। তবে রনির বাবা মায়ের চোখে সেরকম কোনো উৎসাহ চোখে পড়েনি। বড় মেয়ে রিমিকা মাধ্যমিকে সেরকম আহামরি রেজাল্ট করেনি। তবে ইতিহাসে নিরানব্বই পেয়েছিল। রিমিকা বরাবরই ইতিহাস ভালো। ওর ইচ্ছে ইতিহাস নিয়ে মাস্টার্স করবে, গবেষণা করবে। আর্টস নিয়ে নিজের স্কুলে ভর্তি হয়েছিল রিমিকা।

তাই রনি ছিল রিমিকার বড় আদরের। রনি দিদিকে ভীষণ ভালোবাসতো। রনি স্বপ্ন দেখত দিদি ইতিহাস নিয়ে বিদেশে পড়তে যাচ্ছে। দিদি মস্ত বড় ঐতিহাসিক হয়ে দেশে ফিরছে। দিদির স্বপ্নে রনি যেন বিভোর হয়ে থাকত।

তবে সব স্বপ্ন তো উড়বার জন্য মনের মতো আকাশ পায় না। রিমিকা যখন উচ্চমাধ্যমিকে প্রায় নব্বই শতাংশ মার্কস নিয়ে পাস করল, বাবা অর্ধেন্দুবাবু খুব একটা খুশি হলেন না। খুশি না হওয়ার কারণ যদিও অন্য কিছু। মেয়ে জেদ ধরেছে, সে ইতিহাস অনার্স নিয়ে পড়বে। ওদিকে মেয়ের জন্য বিয়ের প্রায় সম্বন্ধ পাকা করে ফেলেছেন তিনি। রেলের চাকরি করা উচ্চশিক্ষিত দাবিহীন পাত্র। এরকম একটা পরিবার, সুযোগ্য পাত্র হাতছাড়া করতে চাইছিলেন না তিনি। বিয়ের পর স্বশুরবাড়ি থেকে বৌমাকে পড়াতে ওদের আপত্তি নেই। কিন্তু সে শুধু ডিগ্রি বাড়ানোর জন্য, নিজের পায়ে দাঁড়ানোর জন্য নয়। অর্থাৎ ইচ্ছের পাখে কুড়ুল মেরে রিমিকাকে বিয়ের পিঁড়িতে বসতে হবে। এটা সে কিছুতেই মেনে নিতে পারল না।

সেদিন সকালের দিকে হালকা বৃষ্টি হয়েছিল। রনি মামার বাড়ি গিয়েছিল। সকালবেলা রনি বাড়ি ফিরে দেখে বাড়িটা অজস্র লোকের ভিড়। ভিড় ঠেলে বাড়ির সদর দরজা পেরিয়ে গেল সে। দেখে তার দিদির নিখর দেহটা সমস্ত খাটের উপর রাখা আছে। হাতের মুঠোয় শক্ত করে ধরা উচ্চমাধ্যমিকের মার্কশিট।

দিদির সঙ্গে সঙ্গে ওর স্বপ্নগুলোকে যেন সমস্ত কেউ ফুল চন্দন দিয়ে সাজিয়ে দিয়েছে। সেদিন রনি মনে মনে প্রতিজ্ঞা করে, "দিদি, তোর স্বপ্নগুলো আমি বাঁচিয়ে রাখব। কথা দিলাম।"

অর্ধেন্দুবাবুর একমাত্র ইচ্ছে, ছেলে ডাক্তারি পড়ুক। এরজন্য জীবনে যত আত্মত্যাগ প্রয়োজন তিনি করবেন। বড় মেয়ের বিবেচনাবুদ্ধিহীন ডিসিশনে তাঁর পরিবার যে বাজে সময়ের মধ্যে দিয়ে গেছে, তার অনেকটা কাটিয়ে উঠেছেন তিনি।

রনির হাতে মার্কশিটটা তুলে দিয়ে অর্ধেন্দুবাবু আবেগঘন গলায় বললেন,
– "বাবা, আমি তোমার রেজাল্টে ভীষণ খুশি হয়েছি। কিন্তু এবার তোমার আসল পরীক্ষা শুরু হবে।"

রনি মুখ নামিয়ে চুপ করে আছে।

– "এবার থেকে অনেক পড়াশোনা করতে হবে। তোমাকে নিট পরীক্ষায় সফল হতে হবে। উচ্চমাধ্যমিক নয়, নিট পরীক্ষাকে পাখির চোখ করতে হবে। আমি আমার অর্জিত সমস্ত কিছু দিয়ে তোমার স্বপ্ন পূরণ করতে চাই।"

রনি মুখ তোলে। এরপর আস্তে গলায় বলে,

– "বাবা, দিদিও চেয়েছিল ওর ইতিহাস পড়ার স্বপ্ন পূরণ হোক। ওর ইচ্ছের দাম দাওনি কেন তুমি?"
অর্ধেন্দুবাবুর মুখটা কেমন যেন ফ্যাকাশে হয়ে গেল। এরপর নিজেকে সামলে নিয়ে বললেন,
– "ইতিহাস পড়ে জীবনে স্ট্যান্ড করা যায় না। ও শুধু ডিগ্রি বাড়ানো ছাড়া আর কিছু নয়।"

রনি এবার বাবার দিকে স্থির তাকিয়ে বলে,

– "এটা তোমাদের ভুল ধারণা। আর তোমাদের এই ভুল আমি ভাঙিয়েই ছাড়ব।"

– "মানে?" অর্ধেন্দুবাবুর চোখেমুখে বিস্ময়।

– "আমার শুধু সায়েন্সের নাম্বারটাই দেখেছ তোমরা। ইতিহাসে যে ৯৯ পেয়েছি সেটা মনে হয় তোমাদের নজরে পড়েনি।"

– "এসব কী বলছিস তুই?" মিনতিদেবী ছেলেকে প্রশ্ন করেন।

রনি মায়ের দিকে ঘুরে বলে,

– "হ্যাঁ, আমি সায়েন্স নয়, আর্টস নিয়ে পড়তে চাই।"

ছেলের কথা শুনে অর্ধেন্দুবাবু আর দাঁড়িয়ে থাকতে পারলেন না।

– "তুই কি পাগল হয়ে গেছিস? তোর বাবার কত স্বপ্ন তাকে নিয়ে, সব ধুলোয় মিশিয়ে দিবি?"

রনি দৃঢ় কণ্ঠে জবাব দেয়,

– "জীবন আমার, তাই সেই জীবনকে নিয়ে আমি কী স্বপ্ন দেখব সেটা আমাকেই ভাবতে দাও মা।"

অর্ধেন্দুবাবু ছেলের মাথায় হাত বুলিয়ে দিয়ে বললেন,

– "তুই অনেক ছোট। এসব ভুলভাল চিন্তা মাথা থেকে ঝেড়ে ফেল। সায়েন্স নিয়ে পড়াশোনা কর। আর্টসে কিছু নেই। জেনারেল লাইনে অনেক কম্পিটিশন।"

রনি কোনো কথার উত্তর না দিয়ে সোজা ঘরে ঢুকে দরজা বন্ধ করে দেয়। দরজা লাগানোর আগে বলে,

– "চিন্তা করো না, দিদির মতো আমি সুইসাইড করব না। তবে ওটাই আমার ফাইনাল ডিসিশন, আমি আর্টস নেব। আর হ্যাঁ, বিষয়টা ইতিহাস নিয়ে পড়তে চাই, যেটা আমার দিদি চেয়েছিল। আর তোমরা সেটা হতে দাওনি।"



দেয়ালে টাঙানো দিদির ছবির সামনে রনি নির্বাক দাঁড়িয়ে রয়েছে। হাতে ধরা রয়েছে মাধ্যমিকের মার্কশিট। ঠিক এই ঘরেই যে ফ্যানটা ঘুরছে, তার নীচে দাঁড়িয়ে গলায় ওড়না পেঁচিয়ে আত্মহত্যা করেছিল রিমিকা। হাতের মুঠোয় শক্ত করে ধরা ছিল উচ্চমাধ্যমিকের মার্কশিট। মুঠোয় মুঠোয় ধরা সেই মার্কশিটের আড়াল থেকে স্পষ্ট দেখা যাচ্ছিল ইতিহাসে ঝলমল করে ওঠা ৯৯ নম্বর। সেদিনই রনি ঠিক করেছিল, দিদির অধরা স্বপ্নের উড়ান যেখানে মুখ খুবড়ে পড়েছিল, সেখান থেকেই তার ইচ্ছের ডানা মেলবে।

দিদির ছবিটা সন্তর্পণে হাতে নেয় রনি। তারপর পকেট থেকে রুমাল বের করে ফ্রেমের উপর জমে থাকা ধুলো মুছে দেয়। ঝকঝকে হয়ে ওঠে ছবিটা। ছবির মধ্যে রিমিকার মুখে ফুটে ওঠা সেই অকৃত্রিম হাসিটা যেন আবার জীবন্ত হয়ে ওঠে। সেই হাসির দিকে তাকিয়ে থাকতে থাকতে রনি ছবিটাকে বুকে জড়িয়ে ধরে।

তারপর কান্না রুদ্ধ গলায় ফিসফিস করে বলে—

"দিদি, আমি ডাক্তার হব না। আমি তোমার মতো মস্ত বড় ঐতিহাসিক হব। তোমার হারিয়ে যাওয়া স্বপ্ন আমি আগলে রেখেছি আমার বুকোর ভেতর। তোমার সমস্ত না-পাওয়াগুলো তোমার এই ভাই একদিন পূরণ করে দেবে।"

চোখ বন্ধ হয়ে আসে রনির। কোথাও যেন রিমিকার এক অদৃশ্য ছায়ামূর্তি, একরাশ হাহাকার আর বেদনা নিয়ে ঘরের চারপাশে এতকাল বন্দি হয়ে ছিল। আজ যেন সে মুক্তি পেল।

মুক্ত রিমিকা যেন তার ভাইয়ের গালে চুমু খেয়ে আনন্দে ঘরময় ছুটে বেড়াচ্ছে। মাঝে মাঝে আদরের ছোট্ট ভাইটাকে জড়িয়ে ধরছে। কোথাও যেন ওর স্বপ্নগুলো আবার রঙিন পাখা মেলে উড়তে প্রস্তুত হচ্ছে।

সমাপ্ত।

– Saptarsha Pal
ME, 3rd Year



– Debraj Chakraborty
ME, 2nd Year



Ode to the Mechanical Mind

We speak in torque and tension,
In blueprints drawn at 2 a.m.
We dream in gears that never sleep,
And shafts that spin without condemn.

Born of fire, shaped by hand,
From Newton's laws to CAD's command.
We bend the steel, we tame the steam,
We turn a sketch into a machine.

From Tesla's coil to Watt's great wheel,
From rocket fins to pistons zeal,
We bridge the gap 'twixt thought and steel
The world moves forward, because we will.

So raise a wrench to entropy,
To chaos turned to symmetry.
This is our craft, our core, our creed:
We build the world the world will need.

– Deb Kumar Das
ME, 1st Year

Beyond the Horizon: Travelling High to get a Self-Driving Adventure

The last week of May had arrived with the familiar heat of Kolkata. The city seemed wrapped in a blanket of humidity, and the roads shimmered under the afternoon sun. Yet, while most people were searching for air-conditioned comfort, my mind was wandering somewhere far beyond the plains—toward snow-capped mountains, deep valleys, winding roads, and endless skies. For months, a dream had been quietly growing inside me. Spiti—that's the name sounded less like places, and more like invitation from the Himalayas themselves. The plan had begun casually over cups of tea with friends. One evening, while discussing our routine lives, someone mentioned, "Why don't we drive to Spiti this year?" Everyone laughed initially. "Kolkata to Spiti? By car?" "It is almost two thousand kilometers one way!" "Who will drive?" The answer came almost immediately. "My Jimny will." The moment those words were spoken, the dream became a mission. My car had already proven itself on rough roads like Ladakh and countryside trips. Compact, sturdy, reliable, and eager for adventure, it seemed almost excited whenever mountains were mentioned. Over the next few weeks, preparations began. Maps were studied. Routes were discussed. Hotels and homestays were shortlisted. The route was ambitious. Nearly two weeks. Countless memories are waiting to happen. The soft hum sounded like the beginning of a grand adventure. As we crossed the Howrah region and joined the highway, the first rays of sunlight appeared behind us. Kolkata slowly disappeared. The first few days were dedicated mostly to covering distance. Highways stretched endlessly across West Bengal, Jharkhand, Uttar Pradesh, and Haryana. We passed green fields, industrial towns, rivers, villages, and bustling cities. At roadside dhabas, we enjoyed hot hand made Chapatis dripping with butter. Every evening brought a new town and a new hotel room. Yet everyone knew that the real journey would begin only after entering Himachal Pradesh. And that moment finally arrived. The landscape changed dramatically. The flat horizon disappeared. The roads began to curve. Mountains appeared like giant guardians standing in the distance. Pine forests emerged. Cool air replaced the summer heat. Everyone became silent for a while. We simply watched. The Himalayas had finally appeared. The Jimny climbed steadily along the winding roads. Every turn revealed another breathtaking view. Clouds drifted across mountain slopes. Villages clung impossibly to steep hillsides. Rivers rushed through deep gorges below. The mountains were not merely scenery. They felt alive. Our first major halt in the mountains was Narkanda, situated at over 8,000 ft.; it welcomed us with cool winds and spectacular views. The evening sky turned golden as the sun disappeared behind distant ridges. We walked through quiet roads lined with cedar trees. A local shopkeeper served us steaming cups of tea saying, "Roads are open now. Snow is melting. You are lucky." His words increased our excitement. That night, standing on a balcony overlooking endless mountains, I realized that the journey had already become special. And we had not even reached Kinnaur yet.



The road toward Kinnaur is one of the most spectacular routes in India. Soon we began driving beside the mighty Sutlej River. The river roared below, carving its way through gigantic mountains. Roads hung precariously on cliffs. At several places, tunnels cut directly through mountains. Massive rock faces towered above us. The scenery became increasingly dramatic. As we moved deeper into Kinnaur, the mountains changed colour. Green forests slowly gave way to rocky landscapes. The terrain looked wilder and more mysterious. The Himalayas seemed to reveal a different personality. By afternoon we reached Sangla Valley. The sight left us speechless. Snow-covered peaks surrounded lush green meadows. Wooden houses stood beside crystal-clear streams. Apple orchards stretched into the distance. The valley looked like a painting. The warmth of the food matched the warmth of the people. Later that evening, we sat outside beneath a sky filled with stars. Far away from city lights, the Milky Way became visible. Thousands of stars illuminated the heavens. Often called the last inhabited village near the Indo-Tibetan border, the road to Chitkul was narrow but incredibly beautiful. Wild flowers bloomed beside mountain streams. Fresh mountain air filled our lungs. Upon arrival, we immediately understood why travellers fall in love with this place. The Baspa River flowed through the valley like liquid crystal. Wooden houses reflected traditional Himalayan architecture. We walked along the riverbank. The sound of rushing water echoed through the valley. Snow sparkled on distant peaks. The sky looked impossibly blue. One of my friends quietly said, "If heaven exists somewhere on earth, perhaps it looks like this." Travel is not only about landscapes. It is also about flavours. And Kinnaur offered plenty. Our next destination was Kalpa. The village faces the magnificent Kinnaur Kailash range. As evening approached, clouds drifted away. Suddenly, the mountain emerged in full glory. Gigantic snow-covered peaks rose directly across the valley. The setting sun painted them golden. We stood watching for nearly an hour. Only a few days earlier, we had been trapped in urban routines. Now we were living among some of the most beautiful landscapes on Earth. Yet the greatest adventure still lay ahead. Beyond Kalpa awaited Nako; Nako lay Tabo; Tabo lay Kaza. And beyond Kaza stretched the legendary land of Spiti. The Cold Desert. The morning in Kalpa began with silence. Not the silence of an empty room, but the silence of mountains. The Kinnaur Kailash range stood before me, glowing faintly under the first light of dawn. The snow-covered peaks gradually transformed from grey to silver and then to gold as the sun climbed higher. The cold mountain air mixed with the aroma of freshly brewed tea, creating one of those perfect travel moments that remain in memory forever. Today we would leave the green valleys of Kinnaur and enter the mysterious world of Spiti. After breakfast, we packed our bags and started the Jimny. The road climbed steadily. The landscape began changing almost immediately. The green forests of Kinnaur slowly disappeared. Trees became fewer. The mountains became larger. The valleys became deeper. The terrain appeared harsher. Road conditions varied constantly. At one moment, we were driving on smooth asphalt.



A few kilometers later, the road transformed into broken gravel scattered with rocks. As we drove higher, the scenery started resembling another planet. Huge mountains of brown, grey, and orange rose around us. Layers of geological history were visible on every cliff. Millions of years of Earth's evolution stood exposed before our eyes. Located at an altitude of around 12,000 feet, Nako seemed suspended between earth and sky. The village surrounded a beautiful lake that reflected the mountains like a mirror. Whitewashed houses, prayer wheels, and Buddhist monasteries gave the place a peaceful atmosphere. Life moved slowly here. Very slowly. Nobody seemed in a hurry. We parked near the lake and began exploring. The water was crystal clear. A gentle breeze created small ripples that distorted the reflections into moving paintings. An elderly monk greeted us near the monastery. He spoke softly and welcomed us warmly. "Long journey," he said, knowing our destiny started from Kolkata. "Very long," then he smiled, "but mountains reward patience." Those words stayed with me. The mountains indeed reward patience. Nothing worthwhile in the Himalayas comes quickly. The next morning, we entered Spiti Valley. Unlike the green beauty of Kinnaur, Spiti possessed a rugged, raw charm. It felt ancient. Timeless. Untouched. The Spiti River flowed beside the road like a ribbon of silver cutting through the desert mountains. The contrast was remarkable. Water was creating life amidst an otherwise harsh environment. Small villages appeared occasionally, looking like tiny islands of civilization surrounded by vast wilderness. Known for its thousand-year-old monastery, Tabo is one of the most important Buddhist centers in the Himalayas. From the outside, the monastery looked simple. Almost humble. But inside, it revealed extraordinary treasures. The monastery carried an atmosphere of deep spirituality. On our way toward Kaza, we made a detour to the ancient Dhankar Monastery, sits dramatically on a cliff overlooking the confluence of the Spiti and Pin Rivers. The monastery appeared to grow directly out of the mountain. Its mud-colored walls blended perfectly with the surrounding cliffs. From the viewpoint, the landscape seemed endless. By evening, we finally arrived in Kaza, the largest town in Spiti; we were now living above 12,000 ft. The following day was dedicated to exploring the famous high-altitude villages around Kaza, the village famous for marine fossils, Langza. Millions of years ago, this entire region lay beneath an ancient ocean. Today, fossils can still be found embedded in rocks. Standing at over 14,000 feet and imagining an ocean beneath our feet challenged the imagination. A giant Buddha statue overlooked the village. Behind it rose majestic snow-covered peaks. Our next stop was Hikkim, home to one of the world's highest post office. The tiny building stood proudly against the backdrop of giant mountains. Naturally, everyone decided to send postcards home. Writing letters in the digital age felt surprisingly meaningful. We experienced the same with an unexpected huge snowfall at the sight; enjoyed a special welcome by the highest valley, wherefrom we continued to Komic, one of the highest motorable villages in the world. When we finally arrived, the village seemed almost unreal. Tiny houses. A monastery. Endless mountains. Infinite sky. The next morning sun rose gently over Kaza, painting the barren mountains in shades of gold and amber. After days in Spiti, we had begun adapting to the rhythm of mountain life, the altitude no longer felt intimidating, and the vast landscapes had become familiar companions. Today promised some of the most famous sights of the entire journey. Key Monastery. Guardian of the Valley. Visible from miles away, the monastery stood dramatically on a hilltop overlooking the Spiti River. At first glance, it resembled a medieval fortress. Tiny villages appeared like scattered dots.



Mountains surrounded everything. A short drive from Kibber brought us to one of the engineering marvels of the Himalayas. The Chicham Bridge. Suspended over a deep gorge, the bridge connects Kibber and Chicham villages. The moment we stepped out of the vehicle, we were stunned. The gorge below seemed endless. Hundreds of feet beneath us, a narrow stream flowed through the rocky canyon. The bridge stretched elegantly across the void. Walking across it was both thrilling and humbling. The wind whistled through the canyon. Looking down required courage. After lunch, we began preparing for the next stage of our journey. The route toward Chandratal, a long waiting journey through the true off-road to the marvel peak of our destination. The road conditions can change dramatically. Water crossings, loose gravel, rocky stretches, unpredictable weather. Everything contributes to the adventure. By evening, we reached Losar, the last inhabited village before Kunzum Pass, at over 15,000 ft, the pass marks an important gateway between Spiti and Lahaul. Snow-covered peaks surrounded us in every direction. From Kunzum Pass, a narrow road branched toward Chandratal. Steep drops appeared beside the road. Loose rocks demanded careful navigation. Yet every kilometer increased our anticipation. Finally, we reached the parking area. From there, a short trek led to the Lake of the Moon. The water shimmered in shades of turquoise and blue. Snow-covered mountains reflected perfectly on its surface. The shoreline curved gracefully, resembling a crescent moon. Above us, the sky performed its nightly masterpiece. The Milky Way stretched across the heavens with astonishing clarity. Some places have a strange ability to silence even the most talkative travellers. Chandratal was one of them. We knew we had to leave, but part of us wished we could remain there for days, watching the changing colors of the lake and listening to the whispers of the wind. Eventually, reality returned. We began our return journey toward Manali. Stretching beneath the mighty Rohtang region, we continued toward one of the most beloved destinations in Himachal Pradesh, Kasol, the Valley of the Parvati. Nestled beside the Parvati River, Kasol has earned a reputation as a peaceful mountain retreat. The road followed winding rivers through forests and small villages. The air felt fresh and cool. As we approached Kasol, the scenery became increasingly enchanting. Dense pine forests covered the slopes. Wooden cottages appeared among the trees. Unlike Spiti's dramatic landscapes, Kasol offered a softer beauty. The mountains here were green and welcoming. The forests seemed alive with birdsong. At several points, we stopped to admire the scenery. Different landscapes evoke different emotions. Spiti inspires awe. Kasol inspires peace. Both are equally beautiful. Our final evening in Kasol became one of the most memorable nights of the entire trip. The Parvati River continued its endless journey through the valley. The very next day, we have started our return towards Kolkata after a powerful ending to the two-week Himalayan adventure.

The journey through Spiti awakened within me a deeper confidence to navigate the rough and uncertain roads of life. As I crossed its harsh terrains and breathtaking landscapes, I learned to endure difficulties with courage while savouring the beauty of nature and the richness of each passing moment. Spiti taught me to live as if every day were a precious gift—embracing life wholeheartedly, free from the weight of endless thoughts about what lies ahead.

- Dr. Arijit Kundu

Ass. Prof., ME

Inline Four (I4) Engine: The Backbone of Modern Automotive Engineering

The Inline Four (I4) engine is one of the most widely used internal combustion engine configurations in the automotive industry. Its simple construction, compact size, reliability, and efficient performance have made it the preferred choice for passenger cars, motorcycles, and light commercial vehicles. The arrangement of four cylinders in a single row provides an excellent balance between power output, fuel economy, and manufacturing cost. This article discusses the construction, working principle, balancing characteristics, advantages, limitations, and significance of the Inline Four engine in modern automotive engineering.

Introduction The internal combustion engine has undergone continuous development since its invention, with engineers striving to achieve higher efficiency, improved performance, and greater reliability. Among the various engine configurations, the Inline Four (I4) engine has established itself as the most popular due to its simple design and practical performance. An Inline Four engine consists of four cylinders arranged in a straight line along a common crankshaft. Each cylinder contributes to power generation through the four-stroke combustion cycle. The engine's compact architecture simplifies manufacturing and maintenance while delivering adequate power for a wide range of vehicles. Because of these advantages, the Inline Four engine has become the standard powertrain for most compact and mid-sized automobiles worldwide.

Construction of an Inline Four Engine The Inline Four engine consists of several essential components that work together to convert the chemical energy of fuel into mechanical energy.

Cylinder Block The cylinder block forms the foundation of the engine. It contains four cylinders arranged in a straight line and houses coolant passages, lubrication galleries, and the crankshaft bearings. It is generally manufactured from cast iron or aluminium alloy for strength and heat dissipation.

Cylinder Head Mounted above the cylinder block, the cylinder head contains the intake and exhaust valves, combustion chambers, spark plugs (for petrol engines), and camshaft assembly. It seals the cylinders and directs airflow into and out of the combustion chamber.

Pistons Four pistons reciprocate inside the cylinders. During combustion, expanding gases force the pistons downward, generating useful mechanical work.

Connecting Rods Each piston is connected to the crankshaft through a connecting rod, which transmits the combustion force while allowing smooth reciprocating motion.

Crankshaft The crankshaft converts the reciprocating motion of the pistons into continuous rotary motion, which ultimately drives the vehicle through the transmission system.

Camshaft and Valve Mechanism The camshaft controls the opening and closing of the intake and exhaust valves through cam lobes and valve train components. Accurate valve timing is essential for efficient engine operation.

Working Principle The Inline Four engine commonly operates on the four-stroke Otto cycle (petrol engine) or the Diesel cycle (compression ignition engine).

Each cylinder completes one power cycle in four distinct strokes. **1. Intake Stroke** The intake valve opens while the piston moves downward, drawing air or an air-fuel mixture into the cylinder. **2. Compression Stroke** Both valves remain closed as the piston moves upward, compressing the charge. Compression increases the pressure and temperature of the mixture, preparing it for combustion. **3. Power Stroke** Near the end of the compression stroke, the spark plug ignites the compressed air-fuel mixture in petrol engines, while diesel fuel auto-ignites due to high compression temperature in diesel engines. The resulting high-pressure gases force the piston downward, producing useful work. **4. Exhaust Stroke** The exhaust valve opens as the piston rises, expelling the burnt gases from the cylinder and completing the cycle. Since each cylinder fires at different intervals, the engine delivers nearly continuous power output. The most commonly used firing order for an Inline Four engine is 1-3-4-2, which provides smooth operation and balanced torque delivery. **Advantages** The Inline Four engine offers several engineering advantages: • Compact and space-efficient design. • Simple construction with a single cylinder head. • Lower manufacturing and maintenance costs. • Good fuel economy due to fewer mechanical losses. • Reliable operation with fewer moving components. • Easy accessibility for servicing and repairs. • Adequate power output for most passenger vehicles. • Good compatibility with both naturally aspirated and forced-induction systems. **Limitations** Despite its popularity, the Inline Four engine has certain limitations: • Secondary vibrations reduce smoothness at high engine speeds. • Limited power output compared to engines with six or eight cylinders. • Larger displacement Inline Four engines may require balance shafts to reduce vibration. • Less refined operation than naturally balanced Inline Six or V6 engines. **Conclusion** The Inline Four engine remains one of the most successful and widely adopted engine configurations in automotive engineering. Its straightforward construction, compact dimensions, fuel-efficient operation, and dependable performance make it an ideal choice for a broad range of vehicles. Although it exhibits secondary vibration characteristics, these are effectively managed through modern balancing techniques. The combination of reliability, simplicity, and cost-effectiveness has ensured the Inline Four engine's continued dominance in the automotive industry, making it a fundamental topic in mechanical and automobile engineering.



– Md Imrul Hossain
ME , 2nd Year



Learning Beyond the Classroom: A Glimpse into India's Hydropower Sector at Maithon Hydel Power Station

Engineering is not only about learning theories in classrooms but also about understanding how these concepts are applied in real-world industries. With this objective, a team of seven Mechanical Engineering students from Kalyani Government Engineering College underwent a 21-day vocational training programme at Maithon Hydel Power Station. The training provided us with a valuable opportunity to observe the operation of a hydroelectric power plant, interact with experienced engineers, and gain practical insights into power generation, plant maintenance, safety practices, and industrial operations. This article highlights the key technical knowledge and experiences we gathered during this enriching industrial exposure.

About Maithon Hydel Power Station : Maithon Hydel Power Station is located on the Barakar River near Maithon Dam, at the border of Dhanbad district in Jharkhand and Paschim Bardhaman district in West Bengal. Operated by the Damodar Valley Corporation (DVC), it is one of the significant hydroelectric power stations in eastern India. The power station is notable for being the first underground hydroelectric power station in India and has an installed generating capacity of approximately 63.2 MW. One of the most memorable experiences of our training was the visit to the underground powerhouse on the very first day. Seeing the massive turbines and generators operating beneath the surface was both exciting and insightful. During the visit, we learned that the powerhouse houses three generating units, two of which have a capacity of 20 MW each, while the third has a capacity of 23.2 MW. Observing these machines our first impressions of the plant was the scale and complexity of its infrastructure. Seeing turbines, generators, control panels, and auxiliary systems operating in real time helped us connect classroom concepts with practical engineering applications.

Technical Insights and Learning Outcomes : This training provided us with valuable exposure to the practical aspects of hydroelectric power generation. During our visits to different sections of the plant, we gained an understanding of how the potential energy of stored water is converted into electrical energy through turbines and generators. We observed the operation of turbine-generator units, control systems, switchyard equipment, and various auxiliary systems that ensure the smooth functioning of the power station. One of the key technical insights gained during the training was the importance of coordination between mechanical and electrical components in power generation. We learned about turbine operation, generator excitation, power transmission, and the role of monitoring and protection systems in maintaining reliability and safety.



The interaction with engineers also helped us understand routine maintenance practices and the challenges involved in operating a large-scale power plant. Beyond technical knowledge, the training enhanced our professional skills. Regular interaction with industry professionals improved our communication abilities and provided insights into workplace ethics, responsibility, and industrial safety. Working together throughout the training programme also strengthened our teamwork, coordination, and problem-solving skills. The experience emphasized the importance of discipline, punctuality, and continuous learning in an engineering career. NOTE : In addition to the common training sessions attended by the group, one of our team members Sherya Saha had the opportunity to visit the Electrical Department. This provided valuable exposure to electrical equipment, power distribution systems, control panels, and maintenance practices. The visit offered a broader understanding of the electrical aspects of power plant operation and complemented the technical knowledge gained during the training programme. Conclusion : Our 21 days at Maithon Hydel Power Station were far more than just a vocational training programme—they were a journey from textbooks to turbines, from theories to real-world engineering. The experience gave us a firsthand look at how a large-scale hydroelectric power plant operates and helped us appreciate the technology, precision, and teamwork behind every unit of electricity generated. Beyond the technical knowledge, the training taught us valuable lessons in discipline, professionalism, and collaboration. It allowed us to interact with experienced engineers, learn from their expertise, and gain a clearer perspective on our future roles as engineers.



– Anubhab Roy
ME , 3rd Year



Can a Graph Tell the Whole Story? A Critical Look at Two Stress-Strain Curves of Mild Steel

Every mechanical engineering student can draw the stress-strain curve of mild steel from memory. It is a fundamental rite of passage into the study of how materials deform under load, complete with its unique peaks, sudden drops, and straight initial line. However, an interesting problem happens when you look through different textbooks, use online simulation tools, or study real laboratory test data from a factory: the graphs rarely look the same. In some classroom diagrams, the initial straight part (the elastic region) is drawn as a gentle, easy-to-read 45-degree angle. In other graphs, it looks like a nearly vertical cliff hugging the vertical axis. Some diagrams carefully map out the small jumping behavior between the upper and lower yield points, while others flatten this entire area into a simple, single step. If both types of graphs show the exact same engineering material—standard mild steel—why do they look so structurally different? This observation motivated our team to investigate the engineering facts behind these visual differences, exploring the assumptions, scales, and contexts that change how engineering data is visually shared.

THE CORE DIVERGENCE: SLOPE OF THE ELASTIC REGION The most striking difference between these two graphs is the geometric slope of the elastic line. In engineering theory, Hooke's Law states that the linear slope of this region represents Young's Modulus (E). Mathematically, it is written as: $E = \sigma / \epsilon$ where σ represents engineering stress and ϵ represents engineering strain. For standard mild steel, Young's Modulus is a well-known physical property, typically valued around: $E \approx 200 - 210 \times 10^9 \text{ Pa} = (200 - 210 \text{ GPa})$ Because Young's Modulus is determined by the strong atomic bonds inside the iron crystal matrix, its numerical value does not change from one standard metal sample to another. Why, then, do the visual slopes of Figure 1 and Figure 2 create completely different geometric angles on the page?

DECONSTRUCTING THE VISUAL DISCREPANCY Reason 1: The Illusion of Axis Scaling This is the most critical factor. The visual geometric angle (θ) of a line drawn on a page depends entirely on the scale limits and aspect ratio chosen for the X and Y axes. In a real physical test, mild steel reaches its yield strength at roughly $\sigma = 250 \text{ MPa}$, but it does so at an incredibly small elastic strain—often around $\epsilon = 0.0012 \text{ mm/mm}$ (or 0.12%).

Department of Mechanical Engineering Magazine • Student Forum 1 If we try to calculate the exact geometric line angle on a perfectly uncompressed 1:1 grid scale (where 1 unit of stress visually matches 1 unit of strain), the true mathematical angle would be: $\theta = \tan^{-1}(210 \times 10^9) \approx 89.9999^\circ$ A line drawn at 89.9999 degrees is visually indistinguishable from the vertical Y-axis.



Therefore, the friendly, readable slope in Figure 1 is simply a graphical adjustment made by stretching the horizontal strain axis wide by several levels of magnitude. The visual angle is not the actual material property: only the math hidden behind it yields the true constant. Reason 2: Classroom Presentation vs. Practical Design Value Figure 1 serves as an educational diagram. Its explicit goal is to zoom into the physical transitions, giving engineering students the visual space needed to separate distinct material milestones such as Lüders bands, yield elongation, strain hardening, and localized narrowing (necking). On the other hand, Figure 2 is an engineering chart optimized for quick calculations. Engineers designing structural frames care about explicit numerical design limits: Ultimate Tensile Strength (UTS), Yield Strength, and total failure elongation. They do not need an artificially wide yield region; they need accurate data placement mapped onto a true, balanced analytical space. CALCULATING THE TRUE GEOMETRIC ANGLE The visual slant of a plotted line can be evaluated using standard trigonometry: $\tan \theta = \Delta y / \Delta x$ Because modern software stretches plots automatically to fill computer windows, the visual angle changes whenever the chart boundary is resized. If we look at the scaled limits of Figure 2—where the stress axis reaches up to 450 MPa over a height of about 10 cm, and the strain axis spreads to 0.25 over a width of 10 cm—the line acts completely differently than a textbook graphic. When these real quantities are plotted without manual distortions, $\theta = \tan^{-1} (210) \approx 89.7^\circ$ when normalized to the grid scales. This fully explains why structural software packages always show an almost vertical elastic boundary line. "In engineering, understanding the scale of a graph is often far more important than simply admiring its visual shape." OUR PROFESSIONAL INTERPRETATION & CONCLUSION The stark difference between these two curves delivers an essential lesson that goes beyond basic material science: Engineers must never blindly trust the silhouette of a graph. They must look deeper to understand the boundaries and assumptions behind it. A line on a chart can be stretched, compressed, simplified, or smoothed out to fit a textbook layout or a university syllabus. However, the true physical behavior of the material remains entirely independent of how it is printed. Just as an experienced mechanical engineer looks past the shiny exterior of a car body to evaluate the thermodynamic engine cycle running inside, we must look past the superficial geometric look of a visual plot to isolate the genuine engineering constants. In conclusion, the two stress-strain curves of mild steel look completely distinct at first glance, especially along the early elastic path. Yet, this visual divergence does not mean the material is behaving differently. It is simply the result of different axis configurations, presentation goals, and communication contexts. Our brief study reminds us that a genuine engineering education is not about memorizing static shapes on a page—it is about reading them with absolute technical accuracy. The true fingerprint of mild steel lies embedded deep within its measured physical properties—its 210 GPa Young's modulus, its yield limit, its ultimate strength, and its ductility—never in the changing visual angle of a printed line.

– Apurba Diyasi & Victor Gorai
ME , 2nd Year

Rocket Design Technology

Rocket science has always been one of the most exciting fields of engineering, combining physics, mathematics, design, and technology to make space exploration possible. Rocket Design Technology provides an introduction to the concepts and engineering principles behind the design and operation of rockets and launch vehicles.

The study of rocket technology begins with understanding how rockets work. Unlike conventional aircraft, rockets carry both fuel and oxidizer, allowing them to operate even in the vacuum of space. The principle behind rocket propulsion is based on Newton's Third Law of Motion, which states that every action has an equal and opposite reaction. The high-speed gases expelled from a rocket engine generate thrust, enabling the vehicle to overcome gravity and travel into space.

An important part of rocket technology is understanding orbital mechanics. Satellites and spacecraft are placed in different types of orbits depending on their mission requirements. These include Low Earth Orbit (LEO), Medium Earth Orbit (MEO), Geostationary Orbit (GEO), and Sun-Synchronous Orbit (SSO).

Learning about these orbits helps in understanding how communication satellites, navigation systems, weather satellites, and Earth observation missions function.

Rocket propulsion systems. Chemical rockets, which include solid, liquid, and hybrid propulsion systems, are widely used in modern launch vehicles. In addition, advanced propulsion technologies such as electric and nuclear propulsion offer new possibilities for long-duration space missions. These systems provide insight into how engineers select propulsion methods based on performance, efficiency, and mission objectives.

Rocket propellants. Different propellant combinations have unique characteristics that affect the performance of a launch vehicle.

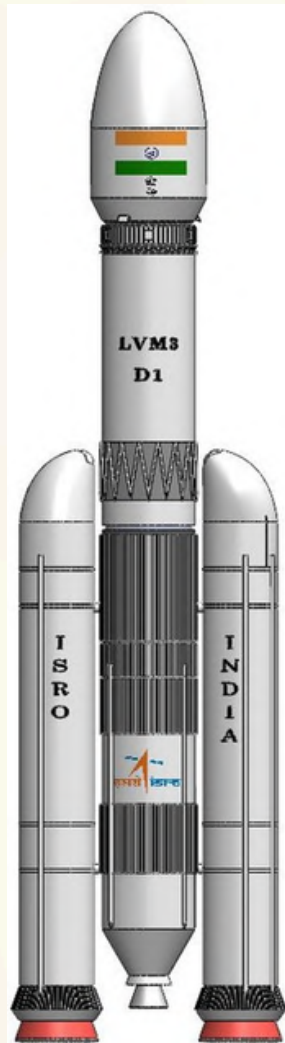
Concepts such as cryogenic, semi-cryogenic, hypergolic, and non-hypergolic propellants demonstrate the complexity involved in rocket design and operation. Understanding these propellants highlights the importance of efficiency, safety, and storage requirements in aerospace engineering.

Rocket anatomy is equally important in understanding the complete structure of a launch vehicle. Components such as the payload fairing, propellant tanks, rocket engines, stage separation systems, and guidance systems work together to ensure a successful mission. Each component is designed to perform a specific function while withstanding the extreme conditions experienced during launch and flight.

Rocket Design Technology not only introduces the fundamentals of space systems but also develops problem-solving, analytical thinking, and design capabilities. The combination of theoretical knowledge and practical modeling experience provides a strong foundation for students interested in aerospace engineering and space technology.

From understanding propulsion systems and orbital mechanics to exploring rocket structures and developing 3D models in SolidWorks. The experience of modeling the LVM3 launch vehicle further demonstrates how

engineering concepts can be applied to real-world aerospace systems, making the learning process both informative and engaging.



- Md Imrul Hossain
ME , 2nd Year

Suspension System

The suspension system is an important part of a vehicle. It is located between the wheels and the vehicle body. The main purpose of the suspension system is to provide a smooth and comfortable ride by absorbing shocks and vibrations from the road. When a vehicle moves over bumps, potholes, or rough roads, the suspension system reduces the impact. This helps passengers feel less discomfort and protects the vehicle from damage. It also keeps the tires in contact with the road, which improves stability, steering, and braking performance. A suspension system mainly consists of springs, shock absorbers (dampers), control arms, and linkages. Springs absorb the energy from road bumps, while shock absorbers control the movement of the springs and prevent excessive bouncing. There are different types of suspension:-

1. **Dependent Suspension:** Both wheels are connected by a rigid axle. Movement of one wheel affects the other. Used in trucks and buses.
2. **Independent Suspension:** Each wheel moves independently. Provides better comfort and handling. Commonly used in cars.
3. **Semi-Independent Suspension:** A combination of dependent and independent suspension. Common in small passenger cars.
4. **Air Suspension:** Uses air springs instead of metal springs. Gives a smoother ride and adjustable height.
5. **Hydraulic Suspension:** Uses hydraulic fluid to absorb shocks and control vehicle height.
6. **Active Suspension:** Electronically controlled suspension that automatically adjusts according to road conditions for maximum comfort and stability.

In conclusion, the suspension system plays a vital role in vehicle safety, comfort, and performance. A well-maintained suspension system ensures a smooth ride, better control, and longer vehicle life.

- Akash Pramanik
ME, 2nd Year

THE SIDE EFFECTS OF BLEEDING

It Starts Quietly.

Bleeding? Yes, some people bleed from bruises, some from injuries, some through nosebleeds. But some people bleed internally, too.

I am not talking about internal haemorrhage. I am talking about the people who devote every minute of their lives towards uncertainty – spending days, weeks, months, and even years chasing their dreams, whatever those dreams might be.

"Relaxing feels abnormal."

1 For them, wasting even a single minute feels frustrating. Not because they are impatient, but because their minds automatically calculate the cost of every passing minute.

SIDE EFFECTS: Relaxing feels abnormal. Bathing feels time-consuming. Even having meals sometimes feels like wasted time.

2 They become obsessed with productivity, work, ideas, possibilities, and understanding how things function. They constantly wonder how people achieve their passion, how systems work, and how life could be improved. Every successful person feels like an idol to learn something new from them.

SIDE EFFECTS: Typical people, communities, and institutions often fail to provide them with grades or to understand them. Their abilities get underestimated. Their unconventional thinking gets misunderstood.

3 They often feel different from almost everyone around their age, even when they cannot properly explain why.

SIDE EFFECTS: They slowly become isolated. People mistake them for introverts, rude individuals, or even useless people, without ever understanding the battles inside their minds.

And perhaps the biggest side effect is this: After being constantly pressured, misunderstood, underestimated, and emotionally exhausted, they slowly begin doubting themselves, too. The same people who once dreamed fearlessly begin questioning their own worth. Life starts feeling worthless when no one on earth understands their battles and keeps questioning their achievements, and never keeps their eyes on their daily battles.

"Eventually, chasing passion starts feeling like trying to touch the horizon where the sea meets the sky – while standing alone in a small boat."

Perhaps the truth is: "A problem-free life costs problems, a truly problem-free life doesn't exist"

Respect to the dreamers.

- Soumyadeep Nandy

ME , 1st Year

Welding Cladding: Some Perspectives

Abstract

Protecting a structure or component from adverse corrosive atmosphere can be done by providing a cladding layer on it. Clad layer tries to resist corrosion and improves service life. Among various cladding processes, weld cladding is widely employed in industry. Usually low carbon steel, commonly known as mild steel (MS), is utilized to manufacture easily several mechanical, civil, chemical structures and components for its advantageous mechanical properties. However, MS can be corroded to a large extent under harsh environmental condition. Traditionally, stainless steel is employed to cover MS base material to retain service life of the structure or component. Many research works are being conducted worldwide to improve cladding procedure for covering suitable stainless steel and/or other anti-corrosive materials onto MS. In this article, a brief overview of the works done is made and an outline on the research works carried out at the Department of Mechanical Engineering of Kalyani Govt. Engineering College is included.

Keywords: Welding, cladding, GMAW, FCAW, GTAW, corrosion resistance.

Introduction Materials surviving severe atmospheric condition for a considerable length of time are often used in some industries; however, they are experienced to be much costly if employed to constitute the whole body of a component. Cheap alternative to it is to cover some low grade cheap materials by a corrosion resistant material to improve service life [1, 2]. Different methods can be resorted to for serving this purpose such as coating, plating, buttering, cladding, hardfacing, etc. In surface coating, few microns-thick layer is deposited on an object. An example is to provide 2 to 15 μm thick coating on a cutting tool to have improved performance. Sprayed coating of 75 to 180 μm thickness is applied to have adequate corrosion resistance. Thermal spraying is used to improve properties of spray-coated metals, alloys, composites and ceramics applied as biomedical coatings, thermal barrier coatings, wear resistant coatings, corrosion resistant coatings and wear resistant coatings engraved with laser beams. Plating is another popular metal deposition method by means of electrolysis, but it can be applied to only electrically conductive materials, in general. Chromium and nickel plating are commonly used in plastic injection moulds made of steel, etc. [2]. Buttering is another process in which a layer is formed on the substrate surface to facilitate proper bonding of weld overlay, otherwise, bonding between the two base materials is not good. In case of mild steel and stainless steel joining, one buttering paste made of nickel powder, ferro-vanadium and ferro-titanium powders was deposited onto the surfaces to join properly with gas tungsten arc welding (GTAW) [1, 2]. In cladding, unlike coating, several millimeters of a corrosion-resistant material is deposited on a corrosion-prone material. It does not change the microstructure of the base material as this process creates a new surface layer with a different composition than that of base material. In general, cladding layer is harder than the base material. Compared with other techniques used for surface treatment by means of material deposition, cladding has some distinct advantages. It provides high hardness, corrosion and/or erosion resistance, good bonding and favourable microstructure [1-10]. When cladding is solely for improving hardness of the surface, it is called hardfacing. It increases abrasive wear resistance. In the present paper, a brief overview of the works done in the area of cladding is made and an outline on the research works carried out at the Department of Mechanical Engineering of Kalyani Govt. Engineering College is presented.

Various Methods of Cladding : Cladding can be done by several processes like rolling, strip cladding, different types of welding, viz. electric resistance welding, shielded metal arc welding (SMAW), electroslag welding (ESW), gas tungsten arc welding (GTAW), flux cored arc welding (FCAW), gas metal arc welding (GMAW),

laser beam welding (LBW), oxyacetylene gas welding, powder welding by laser, pulsed gas metal arc welding process (GMAW-P), tubular core covered electrodes, plasma process (Hot-wire), submerged arc welding process (SAW) (submerged arc strip cladding and strip ESW surfacing), friction cladding, laser engineering net shaping (LENS), explosive welding, etc. Cladding by explosive welding uses an explosive detonation as the energy source to produce a bond between metal components. It can be used to join virtually any metal combination that are metallurgically compatible and also that are non-weldable by conventional processes. 316L stainless steel and DIN-P355GH grade vessel steel were clad by explosive welding technique successfully. Difficult to weld material combinations such as W/Cu, Mo/Cu, Mg/Al, Ti/ stainless steel, etc. were also reported to be welded by explosive welding using underwater shock wave technique. Cladding by rolling is also a popular process and more than 90% of worldwide clad plate production relates to hot roll-bonding [1-10]. Weld cladding is applied widely in chemical industry, fertilizer plants, aviation, mining, agriculture, power generation, food processing, etc. as a cost effective engineering solution. Weld overlay cladding techniques were originally developed for applying on Defense (Navy) components subjected to extreme pressure and shock loading. These clad components come in contact with sea water, but needs less maintenance. A number of parts of the submarine pressure hull are clad with Inconel 625. Pressure vessels used in power plants are reportedly clad to provide anti-corrosive as well as strong surface layer to survive severe working conditions at elevated temperature. Cladding was also tried to provide for enhanced performance of components in service, or to repair worn or corroded components. Large worn out gears could be preliminarily repaired by depositing metallic materials using a cladding technique [1-10]. Among different cladding techniques, laser coating/cladding can be done on shafts, rods and seals, valve parts, sliding valves and discs, exhaust valves in engines, cylinders and rolls, pump components, turbine components, wear plates, sealing joints and joint surfaces, tools, blades, moulds, etc. These clad rods and rolls are used in paper, textile and food industries. Clad tools can be used as cutting, punching or die tools. A new popular technique, namely, laser direct metal deposition (LDMD) is an additive technique based upon the mechanism of fusing metallic powders delivered by either a lateral or coaxial nozzle to a base metal or substrate to build three-dimensional objects directly from a computer aided design (CAD) model. Applications of the laser direct metal deposition process have been found in the aerospace and automotive industries for repair and tooling [4, 5]. Cladding by explosive welding was reported to be used for wide flat plates. The same process was also used for the manufacture of concentrically bonded tubes and pipes. It was reported that Nickel-cladding of steel pipes could well be done by GMAW, GTAW, SMAW or MMAW methods [2]. Nitrogen addition could also give high better corrosion resistance [11, 12]. Pulsed GMAW overlay cladding was done in various components in power plants, especially, in case of water tube boilers. Different types of rolls such as continuous casting rolls, etc. were clad successfully by submerge arc welding, while cold rolls were reported to have been clad by continuous pouring process [2].

Weld Cladding Done Using GMAW at KGEC, Kalyani To safeguard structures exposed to corrosive environment, arc-based weld cladding is a widely adopted technique. Cladding is fundamentally a dissimilar metal welding. Although solid-state welding processes are there, different arc welding techniques are preferred for cladding, with Gas Metal Arc Welding (GMAW) proving particularly effective [1, 3, 8-10]. Several investigations were performed at the Welding Shop of the Department of Mechanical Engineering of Kalyani Govt. Engineering College, Kalyani. While Bhaskar Chakraborty did [13] weld cladding experiments at Welding Technology Centre at Jadavpur University in his M.Tech thesis work using GMAW. a B.Tech group of students led by Khara [14] and Anup Kumar Verma, an M.Tech (PE) student, did [15, 16] the first series of weld cladding experiments at KGEC, Kalyani. Manas Kumar Saha later did Ph.D by investigating on cladding using GMAW [17] and FCAW [18-20] processes. He tried to find out the influence of variables on metallurgical structure, performance and corrosion resistance of clad sections.

Proper weld bead geometry was tried to find out initially through bead-on-plate welding to explore the optimal situation [1, 2, 21]. 2 An ESAB, India manufactured AutoK400 MIG / MAG welding set up was used in the weld cladding work involving GMAW and FCAW processes as shown in Fig. 1. Pure CO₂ gas is employed for shielding the weld region in most of the cases. A mixture of argon and carbon dioxide was also tried in some work. Welding torch has been fitted on a speed controllable vehicle. Straight linear motion is obtained as it follows a guided path without providing any weaving motion. Velocity of it can be set at different values within a range. Shielding gas discharge is set within 10-18 L/min. Low carbon steel base plates have been clad in single layer with 50% overlap. Weld voltage and weld current are chosen suitably to determine heat input (HI) following standard relationship. Austenitic Stainless Steel is taken as the clad material in most of the works [1, 2].

From the micrographs in Fig. 2, it can be observed that small size corrosion pits are there on the clad surface. However, large-scale corrosion can be visible in Fig. 2(b) on the unclad sample. From this Fig. 2, thus, it can be stated that cladding could improve corrosion resistance to a large extent. Cladding is being successfully utilized for structural restoration. Numerous researchers studied bead morphology including Kannan and Yoganandh, who developed [8] a reliable mathematical model to forecast bead profiles based on GMAW settings. Successful cladding of austenitic stainless steel onto low-carbon steel was demonstrated across various experimental conditions. Additionally, Palani and Murugan created [7] effective models to anticipate both bead profile and dilution levels in GMAW processes [1-3]. Saha et al. identified that a heat input of 0.49 kJ/mm yielded the lowest corrosion rate, noting that higher heat levels diminish the ferrite phase and degrade resistance. Similarly, Bose and Das found that 0.6 kJ/mm heat input offered significant corrosion protection for stainless steel cladding. Extensive research explored various material depositions to inhibit corrosion, including the integration of nitrogen. Specifically, Das and Saha achieved [21, 22] enhanced protection by cladding nickel-plated medium carbon steel with austenitic stainless steel. While higher heat input can 3 improve impact toughness, it typically reduces corrosion resistance. Conversely, increasing argon content in the shielding gas was shown to improve corrosion resistance done through immersion testing in a chloride environment.

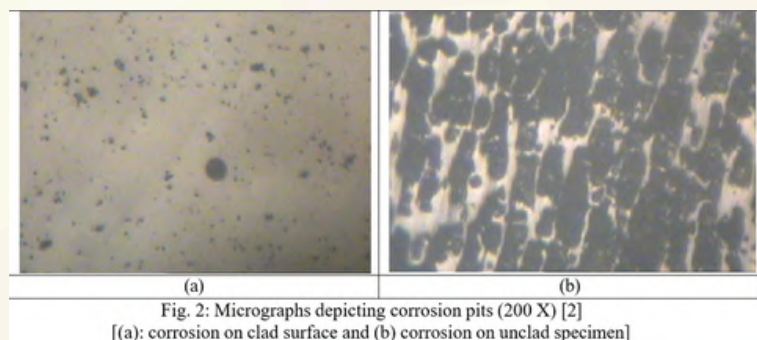
Weld Cladding Done Using FCAW Based on the earlier findings, the characteristics of duplex stainless steel cladding using a wire electrode (as described in Table 1) under different heat input values were also taken up expecting superior anti-corrosive performance, first by Bhaskar Chakraborty [13] and then by a team of students led by S. Das and Manas Kumar Saha [17, 23]. The welding process utilized was FCAW (flux-cored arc welding) as the duplex stainless steel wire electrode used had flux material placed inside the core of the wire electrode. Although some works were reported in the literature [24] near to this experimental work, revalidation of the earlier findings and improvement over those was the aim of those works. The specimens were subject to immersion testing in chloride and sulphate environment to identify the optimal parameters for achieving minimum corrosion rate.

The final configuration of the weld bead is determined by heat input selections; for instance, duplex stainless steel was efficiently layered onto low-carbon steel under specific parameters. Saha et al. reported [18-20] high durability using duplex stainless steel fillers in single-layer cladding. Carbon, silicon, manganese, chromium, molybdenum and nickel content of duplex stainless steel filler wire (E2209T0-1) are remarkably higher than that of the base plate. The electrode wire contains nitrogen (0.125%) to boost precipitation of austenitic phase and also to stabilize it, thus improving mechanical properties of stainless steels. Ferrite phase of this steel offers better corrosion resistance than the γ -phase. Additionally, Kannan and Murugan also observed [24] that rising weld current led to increased dilution, penetration, reinforcement and bead width. Previous works generally indicated that duplex stainless steels outperformed austenitic grades in corrosive environment [1, 2].

Applying a Buttering Layer to Promote Cladding Application of a buttering layer on the substrate and cladding a superior material on the same were found to promote cladding performance. Saha did two types of buttering layers with nickel and copper coating [21, 22] on MS samples and austenitic stainless steel cladding was done on them to report better anti-corrosive characteristics. Weld Cladding Done Using GTAW Gas tungsten arc welding (GTAW), commonly known as tungsten insert gas (TIG) welding, makes welded joint with good appearance and high quality. However, low productivity of GTAW limits its wide application. Again, pulsed GTAW (GTAW-P) or pulsed TIG (P-TIG) welding was also tried to improve productivity of GTAW and also for cladding [25, 26]. There are several categories of stainless steel based on crystal structure such as ferritic stainless steel (BCC structure), martensitic stainless steel, austenitic stainless steel (FCC structure), etc. and some of them are utilized for cladding MS objects. Beside these, in some cases, duplex stainless steel which is a mixture of 50% Ferrite & 50% austenitic stainless steels were also tried [16]. Bose and Das utilized pulsed TIG welding to establish 0.594 kJ/mm as the ideal heat input for minimizing decay [25, 26]. Multiple Dissimilar Cladding Using GTAW Recently another work was undertaken considering multiple layer cladding on MS substrate. Ayan Dey first made a single layer stainless steel cladding and then made a GTAW cladding with a layer of Inconel to achieve high resistance to corrosion under chloride environment [27].

Conclusion Following inference can be drawn on the basis of the review work and the observations made through the experimental investigation on corrosion of different types of cladding undertaken on low carbon steel specimens using a variety of arc welding processes:

- 4 □ Increase in heat input is found not to have a clear trend on the corrosion rate of duplex stainless steel cladding made using FCAW process under chloride environment.
- Duplex stainless steel cladding reduces rate of corrosion substantially over that of unclad low carbon steel.
- GTAW-based Inconel cladding on stainless steel has been reported to give superior corrosion resistance.
- A buttering layer of nickel or copper was also observed to have contributed to better corrosion resistance than without it through GMAW cladding.
- Appropriate heat input, shielding gas flow rate and filler material are to be chosen to obtain favourable microstructure of clad surface that would impart desirably high corrosion resistance under some adverse environment. And research works are going on in this direction towards making recommendations for practicing in industry.



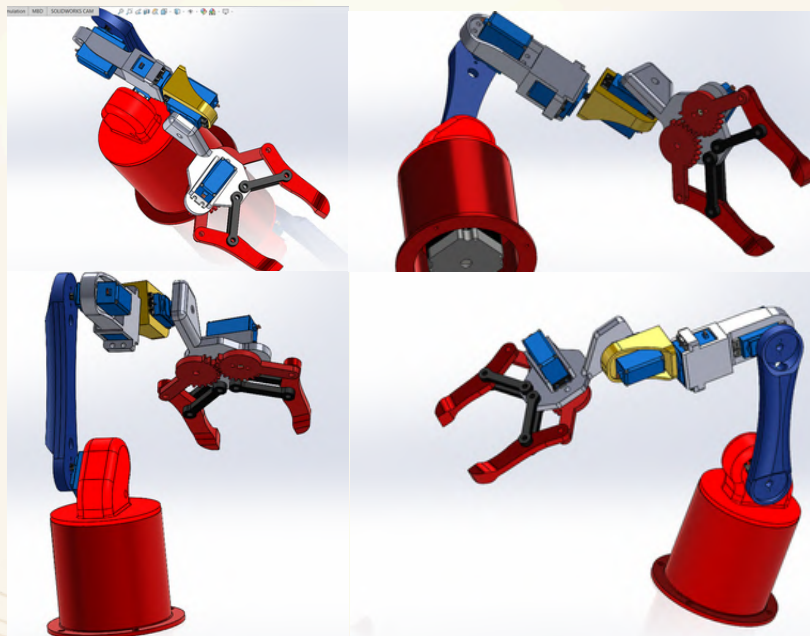
Prof. Dr. Santanu Das
Professor and Former Head, Department of
Mechanical Engineering and Registrar (Officiating),
Kalyani Govt. Engineering College,

6-DOF Robotic Manipulator with Adaptive Gripper

This project presents the design and 3D modeling of a six-degree-of-freedom (6-DOF) robotic manipulator equipped with an adaptive parallel gripper, developed to perform precise pick-and-place and material-handling operations. The robotic arm incorporates multiple rotational joints that provide a high degree of flexibility, enabling it to reach complex positions and orientations within its workspace. The end-effector is designed with a servo-actuated gripper mechanism capable of securely grasping objects of varying shapes and sizes.

The model was developed using advanced CAD techniques, emphasizing structural integrity, smooth motion transmission, and ease of manufacturing. The design demonstrates the practical application of robotics, mechanical design, kinematic principles, and automation technology. Such robotic systems have wide-ranging applications in industrial automation, assembly lines, packaging, inspection, and smart manufacturing environments.

This project reflects the integration of engineering design and innovation, highlighting the growing role of robotics in enhancing productivity, precision, and operational efficiency in modern industries.



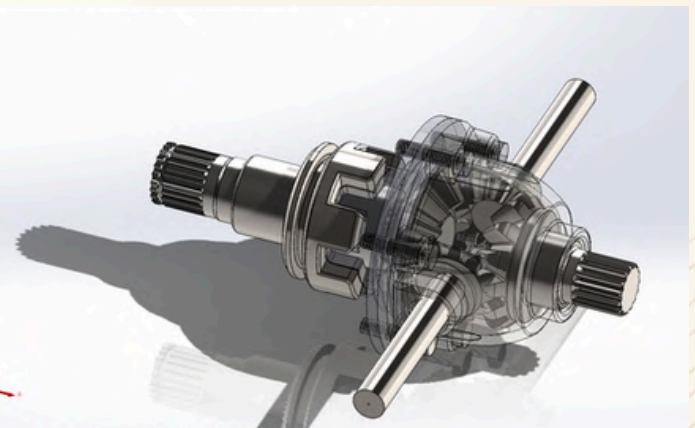
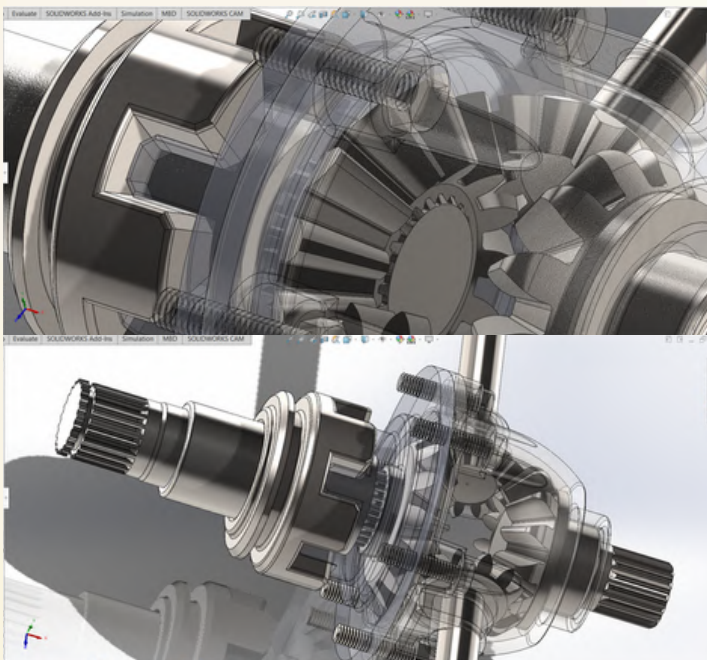
– Atanu Pal
ME, 3rd Year

Automotive Differential Assembly and Motion Simulation

This project showcases the 3D modeling, assembly, and motion simulation of an automotive differential mechanism, developed using SolidWorks. The differential is a critical drivetrain component that enables the drive wheels to rotate at different speeds while transmitting power from the engine, ensuring smooth vehicle operation during cornering and varying road conditions.

The model includes key components such as the differential casing, bevel gears, side gears, pinion gears, shafts, bearings, and fastening elements. Special emphasis was placed on accurate geometric modeling, assembly constraints, and kinematic motion analysis to visualize power transmission within the mechanism. Transparent housing visualization was incorporated to provide a clear understanding of the internal gear interactions and differential action.

This project demonstrates the application of machine design, gear kinematics, CAD modeling, and mechanical system simulation. It serves as an effective educational tool for understanding the working principles of automotive power transmission systems and highlights the importance of virtual prototyping in modern engineering design.



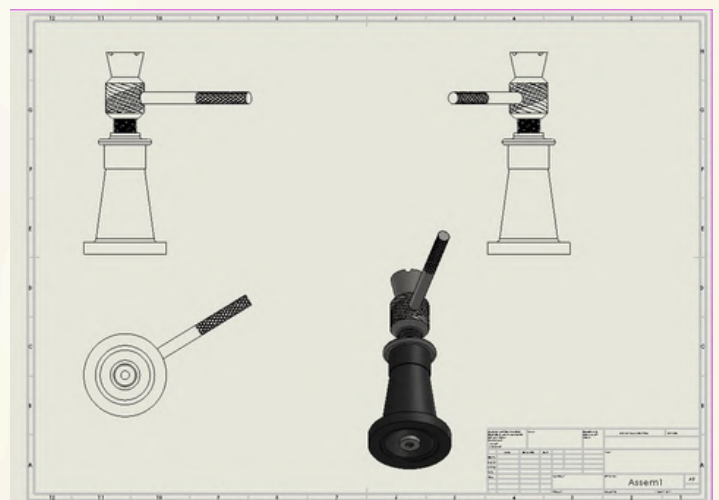
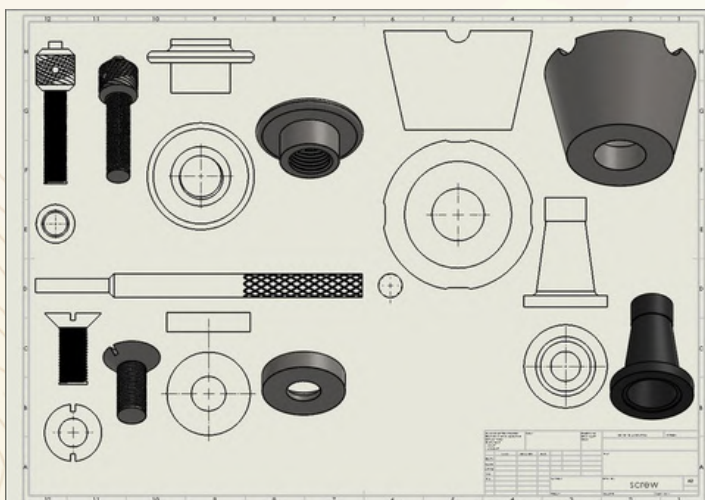
– Atanu Pal
ME, 3rd Year

Screw Jack, used for load lifting

A screw jack is a simple yet highly efficient mechanical device used to lift, support, or lower heavy loads by converting rotary motion into linear motion through the principle of a power screw. It operates based on the law of the screw, where a small turning effort applied to the handle produces a large lifting force due to the mechanical advantage provided by the threaded screw.

The assembly mainly consists of a base, body, threaded screw, nut, and a handle (tommy bar). When the handle is rotated clockwise, the screw moves upward, lifting the load placed on its top. Rotating the handle in the opposite direction lowers the load safely and steadily. The self-locking nature of the screw thread prevents the load from descending on its own, making the screw jack a reliable and safe lifting device.

The model shown represents the assembly of a manually operated screw jack with orthographic views and an isometric view, illustrating the arrangement of its components and the mechanism responsible for load lifting. This assembly demonstrates the practical application of screw threads in mechanical engineering and highlights the importance of power transmission mechanisms in everyday industrial equipment.



– Dipanjan Ghosh
ME, 3rd Year

EXPERIMENTAL INVESTIGATION TO REDUCE THERMAL DAMAGE AND THRUST FORCE IN ORTHOPAEDIC DRILLING

Abstract

Orthopaedic bone drilling is a critical surgical procedure, but manual drilling often results in uncontrolled cutting conditions that generate excessive heat and force. A rapid rise in drilling temperature can cause thermal necrosis of bone tissue (45–47°C), while excessive thrust force may lead to bone cracking or fracture, increasing the risk of surgical complications. These challenges highlight the need for a precise and controlled drilling system.

To address these issues, a customizable automated mini-CNC drilling machine was developed using stepper motors, an Arduino Uno controller, and Universal G-Code Sender (UGS). The system provides accurate control of spindle speed, feed rate, and drilling motion, ensuring stable and repeatable operations. Experimental investigations were conducted on goat femur bone, whose mechanical properties closely resemble those of human cortical bone.

Drilling experiments were performed by varying three process parameters—drill diameter, feed rate, and spindle speed—each at three levels, resulting in nine experimental trials. The responses evaluated were drilling temperature, thrust force, and material removal rate (MRR). Thrust force was measured using a Max-6675 load cell dynamometer, while drilling temperature was recorded with a K-type thermocouple. Experimental results were validated through regression analysis in Minitab software, with prediction errors remaining below 10%, confirming good agreement between measured and predicted values.

Three-dimensional response surface plots revealed that lower drill diameter and feed rate combined with higher spindle speed significantly reduced temperature generation. Similarly, thrust force decreased at higher drill diameters and lower feed rates within the spindle speed range of 20,000–35,000 rpm. Multi-objective optimization using Response Surface Methodology (RSM) identified the optimum drilling parameters as a 3 mm drill diameter, 30 mm/min feed rate, and 32,518 rpm spindle speed, achieving minimum temperature rise and thrust force while improving drilling performance.

Keywords: Bone drilling, Mini-CNC, Thermal necrosis, Thrust force, Response Surface Methodology, Optimization.

– Jaladhar Pal
ME, 3rd Year

DESIGN AND SELECTION OF OPTIMAL TYPE OF HEAT EXCHANGER FOR DOMESTIC USE

Abstract

Here is a refined version that maintains the technical depth and engineering scope, compressed perfectly into a standard single-page abstract length.

Abstract

This technical report presents a systematic engineering framework for the design, evaluation, and selection of an optimal heat exchanger tailored for domestic applications. Driven by the need for residential energy efficiency and sustainable building infrastructure, this study balances thermal performance, fluid dynamics, manufacturing costs, and long-term operational reliability. The research applies the core principles of thermodynamics, fluid mechanics, and psychrometry to evaluate standard heat exchanger topologies—specifically shell-and-tube, plate-and-frame, and finned-tube configurations—under typical household operating conditions such as domestic hot water production and space heating/cooling cycles.

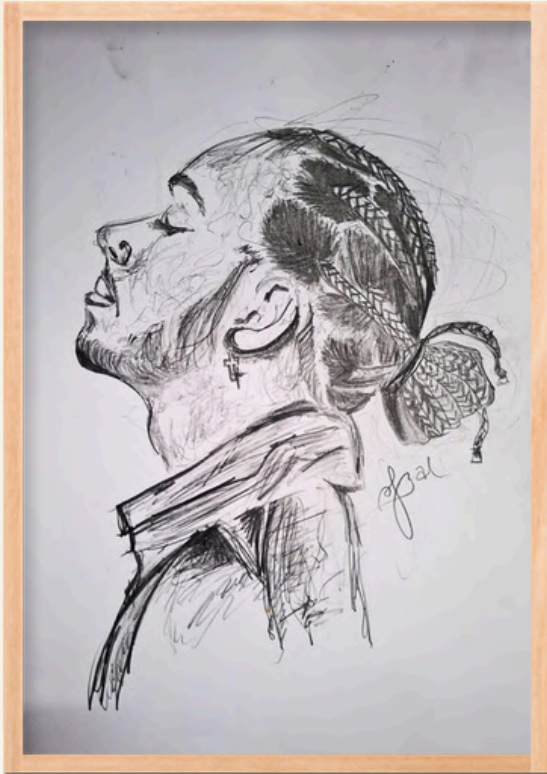
The analytical methodology utilizes the Log Mean Temperature Difference approach and the Effectiveness-Number of Transfer Units method to evaluate thermal performance without relying on overly complex mathematical modeling. Fluid flow configurations, particularly counter-flow versus parallel-flow, are analyzed alongside spatial constraints, hydraulic pressure drops, and subsequent parasitic pumping requirements. Material selection is carefully weighed, balancing the high thermal conductivity of copper and aluminum against the superior corrosion resistance of stainless steel. Furthermore, the study integrates psychrometric analysis to decouple sensible and latent heat loads, ensuring the system accommodates fluctuating ambient humidity, occupant density, and building envelope thermal losses.

Ultimately, this report delivers a multi-criteria decision matrix prioritizing maximum thermal effectiveness, minimal physical footprint, and ease of maintenance. The findings demonstrate that compact brazed plate heat exchangers offer the highest efficiency-to-volume ratio for liquid-to-liquid domestic applications, while finned-tube geometries remain optimal for air-coupled ventilation systems, providing a practical design guide for modern residential thermal management.

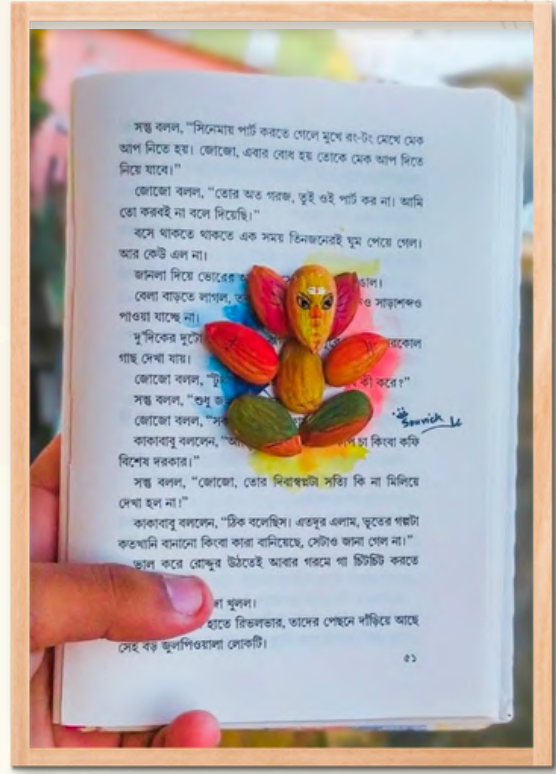
Keywords: Heat Exchanger Design, Residential HVAC, Thermal Effectiveness, Psychrometric Load Analysis, Fluid Dynamics, Domestic Hot Water.

– **Dipanjan Ghosh**
ME, 3rd Year

ARTWORKS



Afzal Hossain Mallick
ME, 4th Year



Souvik ray
ME, 1st Year



Debraj Chakraborty
ME, 2nd Year



Debraj Chakraborty
ME, 2nd Year



Debraj Chakraborty
ME, 2nd Year



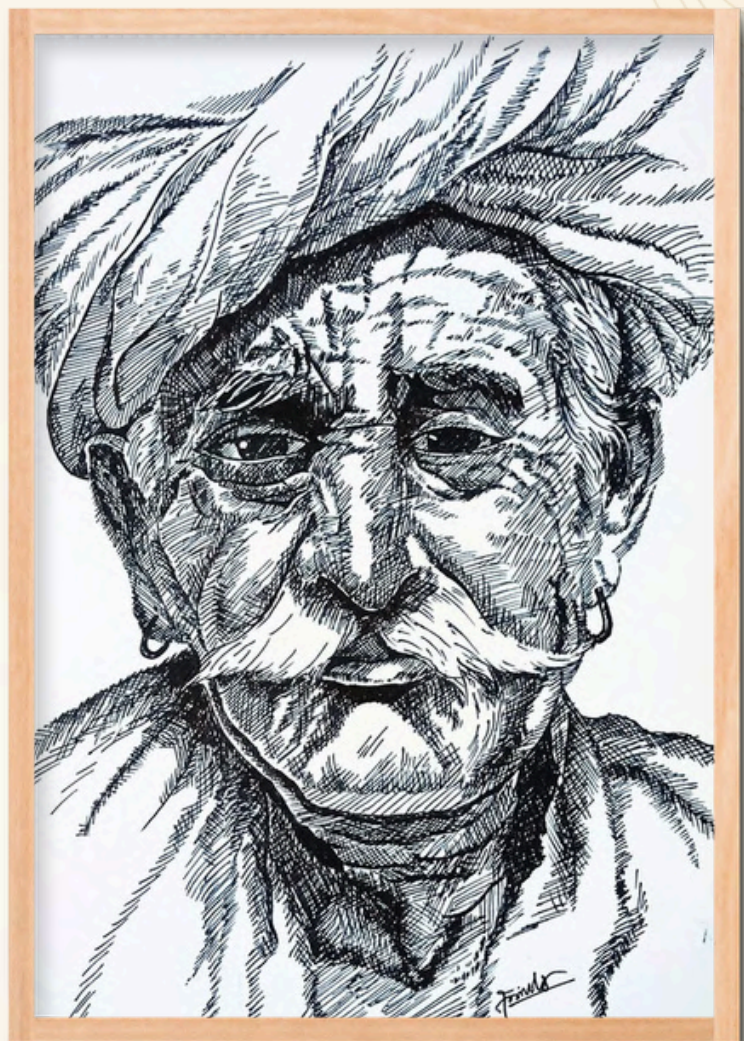
Souvik Ray
ME, 1st Year



Pritam Mondal
ME, 2nd Year



Souvik Ray
ME, 2nd Year



Md Imrul Hossain
ME, 2nd Year



Jatanti Saren
ME, 2nd Year



Sayantan Sarkar
ME, 3rd Year



Sayantan Sarkar
ME, 3rd Year



Md Imrul Hossain
ME, 2nd Year

PHOTOGRAPHS



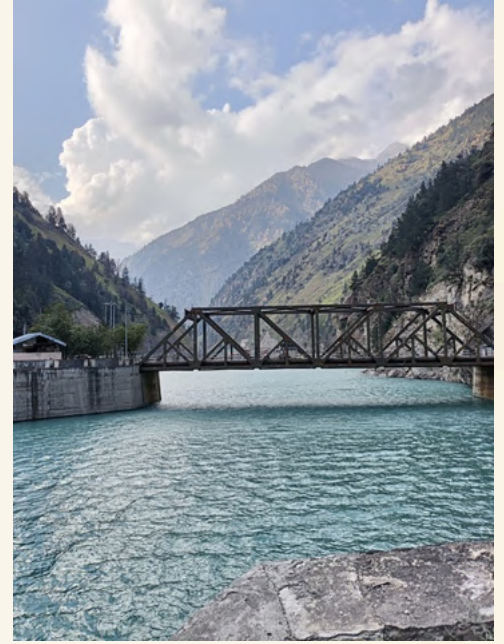
Racktim Das
ME, 3rd Year



Afzal Hossain Mallick
ME, 4th Year



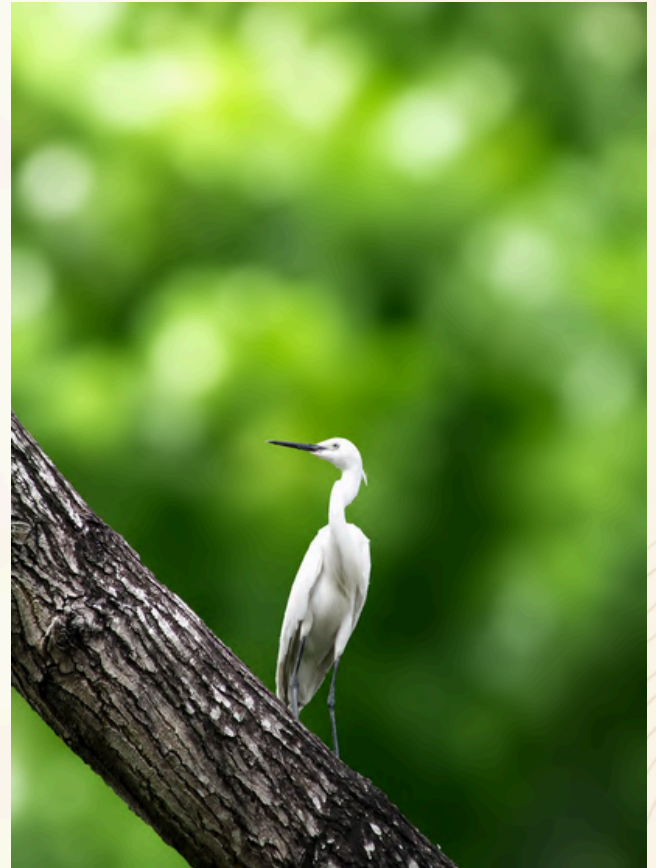
Nayan Biswas
ME, 2nd Year



Sudipto Sardar
ME, 1st Year



Vivek Chel
ME, 1st Year



Nisarga Pal
ME, 2nd Year



Aritra Mukhopadhyay
ME, 3rd Year



Sahil Sheikh
ME, 3rd Year



Ankur Bhattacharya
ME, 3rd Year



THANK YOU

DEPARTMENT OF MECHANICAL ENGINEERING
KALYANI GOVERNMENT ENGINEERING COLLEGE